

How to train your dragon: Evaluating prompting and fine-tuning for GPT-based item generation in L2 listening assessment

Vahid Aryadoust^{*}, Joann Wong

National Institute of Education, Nanyang Technological University, 1 Nanyang Walk, 637616, Singapore

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ABSTRACT

Recent advances in large language models (LLMs), notably GPT models, have introduced new possibilities for automatic item generation (AIG) through prompt engineering. Although iterative prompt refinement can yield measurable improvements, this process eventually plateaus, as outputs remain inconsistent or misaligned with assessment constructs. This study reports an experimental comparison of prompting and fine-tuning to advance AIG for L2 listening assessment. First, we employed prompting and refined the instruction design over three successive iterations to determine an optimised prompt. We then fine-tuned GPT-4.1 using the same optimised prompt, holding prompt design constant to isolate the effect of model adaptation. We generated a total of 40 tests and 240 multiple-choice items in the four model conditions: three prompt-only iterations and one fine-tuned iteration. To contextualize model performance against professional assessment standards, we also compared the GPT-generated items with 245 human-generated listening items used as the fine-tuning dataset. The generated items were evaluated using a hybrid two-tiered framework that combined rule-based metrics with human review to assess content validity and scalability. Fine-tuning produced stronger outcomes overall, yielding items that were more contextually grounded, linguistically coherent, and balanced than those generated through prompting alone, with some requiring minimal or no revision; however, generating higher-order items involving discourse-level reasoning remained challenging. The expert comparison showed that fine-tuned items performed comparably to human-authored items on passage dependence but remained relatively weaker in avoiding absolute language and in targeting localized spans of necessary information. Additionally, issues such as longest-correct-option bias and uneven key distribution persisted, indicating limitations inherent in LLM-generated items. These findings demonstrate the value of fine-tuning for improving item quality and stability, while underscoring the continued need for multidimensional evaluation frameworks and expert benchmarking to ensure construct-aligned, valid assessment design.

1. Introduction

In the era of growing demands for digital assessments, educational institutions require extensive banks of high-quality items to support large-scale online testing and adaptive learning systems (Circi et al., 2023). Traditional item development, however, is slow, resource-intensive, and difficult to scale (Aryadoust et al., 2024; Gierl & Lai, 2013). Automatic item generation (AIG) was introduced to address these challenges through template-based methods that systematically manipulate pre-defined linguistic and cognitive schemas to produce multiple items from a single model (Song et al., 2025). Although effective for scalability, this approach has been criticised for generating items that are repetitive, mechanistic, and insufficiently sensitive to nuanced

language use (Kurdi et al., 2020), resulting in limited linguistic variety and shallow cognitive engagement that fall short of representing the complexity of real-world target language use. In this study, we examine these issues within the specific context of L2 listening assessment.

Recent advances in large language models (LLMs) have transformed AIG by enabling more flexible and context-sensitive item creation. Current LLM-based AIG follows two main approaches: prompt-based generation and supervised fine-tuning, with retrieval-augmented pipelines positioned as hybrid extensions rather than a standalone category. Prompt-based methods rely on task-specific instructions and in-context examples to steer the model's output (Aryadoust et al., 2024; Mucciaccia et al., 2025; Scaria et al., 2024a; Tran et al., 2023), whereas fine-tuning involves further training a pre-trained model on a

^{*} Corresponding author.

E-mail addresses: vahid.aryadoust@nie.edu.sg (V. Aryadoust), qi.wong@nie.edu.sg (J. Wong).

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domain-specific dataset so that it internalises the linguistic, cognitive, and quality criteria required for assessment purposes, rather than relying solely on general-purpose knowledge (Yu et al., 2024). Evidence regarding the effectiveness of prompt-based AIG is mixed. Studies have shown that, despite strong linguistic fluency and rapid generation capabilities, prompt-based items often exhibit inconsistent quality, weak construct alignment, unreliable difficulty control, and limited adherence to principled test-design criteria (Aryadoust et al., 2024; Grevisse, 2024; Hwang et al., 2024; Yao et al., 2025). For instance, Hwang et al. (2024) reported that GPT-4 struggled to generate items at specified difficulty levels, and Grevisse (2024) noted that although linguistic metrics are commonly used, evaluation remains largely disconnected from established multiple-choice design principles. Aryadoust et al. (2024) similarly found that most listening items produced via prompting ChatGPT fell below acceptable quality thresholds. These findings indicate that prompt-based AIG, in its naïve form, may underutilise the adaptive capacity of LLMs for domain-specific assessment tasks.

In contrast, studies applying fine-tuning have reported clearer gains when the model is adapted to the AIG task. Yu et al. (2024) demonstrated that T5 and BART models fine-tuned using retrieval-augmented pretraining and knowledge-graph integration achieved state-of-the-art distractor-generation performance (e.g., F1@3¹ improvements on MCQ and SciQ), outperforming prompt-only baselines. These results suggest that when fine-tuning is appropriately targeted, it can produce more domain-aligned, cognitively coherent, and evaluation-ready items. However, superiority is not strictly one-directional. Bitew et al. (2025) directly compared a fine-tuned multilingual T5 (mT5) distractor generator with a ChatGPT model prompted using dynamically retrieved in-context examples. Teacher evaluations favoured the prompting approach: the Good-Distractor-Rate@10 was approximately 46.7% for the retrieval-augmented ChatGPT versus 24.5% for the fine-tuned mT5 on the language-learning subset, and around 53% of distractors were judged immediately useable. This indicates that well-designed prompting, especially when supported by retrieval of high-quality demonstrations, can outperform a naïvely fine-tuned baseline.

A further challenge lies in the evaluation of AI-generated items, as current frameworks are not psychometrically robust or scalable (Alhazmi et al., 2024). In a systematic review of 100 AIG studies, Alhazmi et al. (2024) argued that widely used automatic metrics adapted from natural language generation (e.g., Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), and the Metric for Evaluation of Translation with Explicit ORdering (METEOR)) capture surface-level lexical similarity rather than educational quality, which can misrepresent distractor validity and item functioning (see Moon et al., 2022; Moore et al., 2023, for a similar discussion). For example, BLEU provides only lexical diversity and does not reflect meaningful differentiation or difficulty (Moon et al., 2022; Raina et al., 2023). As a result, open-domain distractor quality still depends heavily on manual checks, such as “nonsense-distractor rate” (Bitew et al., 2025). Another approach adapted includes using the Item-Writing Flaws (IWFs) rubric, which despite offering a principled, education-grounded tool for evaluating stems and options, its automation and application remains limited and unvalidated for AIG (Moore et al., 2023).

A complete listening assessment comprises multiple interdependent components to reflect authentic listening tasks. At the stimulus level, listening involves processing time-bound acoustic signals that carry phonological, prosodic, and pragmatic information — including features such as speech rate, stress, rhythm, intonation, and speaker variation such as accent — which constitute the auditory input that test-takers must comprehend (Buck, 2001; Field, 2008; Harding, 2018; Nishizawa, 2023; Shin et al., 2021). At the task level, test items

operationalise the construct by eliciting responses that serve as evidence of listening process and skills (Aryadoust & Jia, 2025; Aryadoust & Luo, 2023; In'nami & Koizumi, 2012). These two levels are conceptually distinct: while auditory stimulus design determines the acoustic and phonological demands placed on the listener (Field, 2013), item design determines which aspects of comprehension are targeted and how evidence of that comprehension is elicited (Aryadoust & Luo, 2023). Test development is therefore an inherently multi-stage process, and advances at one stage do not presuppose completion of the others.

Against this backdrop, the present study addresses two key gaps situated within the context of second language (L2) listening assessment, with a particular focus on the item generation stage of test development. Specifically, the study targets lecture comprehension task development (O'Grady, 2024) and task types such as multiple-choice questions (MCQs; Field, 2013; O'Grady, 2021). This focus is deliberate, since, as previous research shows, listening test item generation represents a foundational and resource-intensive stage of the test development pipeline, and improvements here have direct implications for the scalability and quality of assessment design more broadly (Runge et al., 2024). Subsequent stages—including the integration of audio stimuli, the modelling of phonological and prosodic features, and the full psychometric validation of items in operational contexts—are important and complementary (see, for example, Aryadoust & Goh, 2013; Harding, 2018; Nishizawa, 2023) but fall outside the scope of the present investigation. Listening test development remains critically under-researched (Aryadoust, 2017; Aryadoust & Jia, 2025; Aryadoust & Luo, 2023; Goh & Aryadoust, 2025; O'Grady, 2021; Runge et al., 2024), and long-standing concerns persist regarding inadequate research on content representation, construct operationalisation, and principled test development practices in listening assessment (Buck, 2001; Goh & Aryadoust, 2025; also see chapters in Wagner et al., 2024).

Specifically, the study proposes and evaluates a systematic AIG approach that leverages LLMs for item creation, with potential applicability extending beyond listening to other assessment domains. First, the study examines whether fine-tuning can meaningfully overcome the performance plateau observed with optimised prompt-only methods and produce MCQs that more consistently align with construct requirements. Second, it introduces a scalable, two-tiered evaluation framework that integrates human judgement, rule-based checks, and LLM-assisted analyses informed by AIG research (Aryadoust et al., 2024; Grevisse, 2024; Hwang et al., 2024; Yao et al., 2025) and research into listening test design (Aryadoust & Jia, 2025; Aryadoust & Luo, 2023; Buck, 2001; Nishizawa, 2023; O'Grady, 2021, 2024; Shin et al., 2021), enabling a more rigorous assessment of item quality than currently available approaches. To this end, we conducted a multi-phase investigation: Phase 1 refined prompt-engineering strategies for listening item generation, while Phases 2 and 3 assessed the marginal benefit of fine-tuning over prompt-only methods. The following sections review prior work, outline the study design, and present the proposed evaluation framework.

2. Literature review

Automatic item generation (AIG) refers to the use of computational methods to automatically produce test items, offering a potential solution to the cost and labour demands of traditional item development. Creating high-quality items manually is resource-intensive: the cost of developing a single item for high-stakes assessments like the GMAT is in the range of US\$1500 to 2500 (Rudner, 2010), and assembling large item banks requires extensive time and expertise. AIG seeks to address these challenges by generating large pools of high-quality items at scale while significantly reducing development costs (Gierl & Lai, 2012). Early and modern approaches to AIG will be examined in the subsequent sections.

¹ The F1 score (harmonic mean of precision and recall) computed over the top 3 generated distractors.

2.1. Early approaches to AIG

Early approaches to AIG were predominantly template-based, relying on strong or weak theory to guide item design. Mislevy et al.'s (2003) Evidence-Centered Design (ECD) is a useful framework underpinning some of these developments, which emphasises the explicit articulation of claims about student knowledge, the evidence needed to support those claims, and the tasks that can elicit such evidence. ECD provides a conceptual foundation for linking cognitive models to item design in a principled way, thereby ensuring that generated items could be aligned with constructs of interest.

A widely used template method, proposed by Gierl and Lai (2016), involves a three-step process: constructing cognitive models to represent the knowledge and skills to be measured, creating item models to specify the structure of test questions, and then applying algorithms to generate multiple item variants from these models. Building on this foundation, subskill-based approaches to test design represent a concrete operationalisation of cognitive modelling within language assessment (Aryadoust & Luo, 2023). Rather than treating listening ability as a unitary construct, researchers have conceptualised it as comprising multiple interacting subskills, such as global (gist) comprehension, local (detail) recognition, inference-making, pragmatic interpretation (e.g., speaker attitude), and discourse organisation (Aryadoust & Luo, 2023; Buck, 2001; Field, 2008). In this view, item models are not merely structural templates but are explicitly aligned with targeted listening subskills or processes, allowing assessment tasks to systematically elicit evidence for distinct components of the construct. Such subskill alignment strengthens the interpretive warrant between item performance and claims about listening ability as elicited by the test items, consistent with the principles of ECD.

Moreover, principled multiple-choice design serves as an additional layer of construct protection. Haladyna et al. (2022) synthesise evidence-based guidelines for MCQ construction, emphasising distractor plausibility, option independence, clarity of stems, and avoidance of cueing or structural bias. These principles function not only as stylistic conventions but as safeguards against sources of construct-irrelevant variance. When integrated into item modelling, subskill alignment and principled MCQ design together operationalise ECD's claims-evidence-task framework in a manner that is directly applicable to language and listening assessment contexts.

While template-based AIG has proven effective in generating large volumes of assessment items, it has several notable limitations. First, a systematic review by Kurdi et al. (2020) found that items produced are often simplistic and tend to target only lower levels of cognitive ability. Their reliance on rigid, rule-driven structures further restricts flexibility, making it difficult to adapt such methods across domains or to support more complex item types (Maity et al., 2024). Second, although template-based approaches are motivated by the need to reduce costs, the creation of cognitive models and item templates is itself a demanding and resource-intensive process. As described by Pugh et al. (2016), the process of developing cognitive models is rigorous and requires input from multiple subject matter experts. For instance, template design typically involves either analysing existing questions by hand or consulting with domain specialists—both of which are time-consuming and costly (Kurdi et al., 2020). This undermines the scalability of template-based AIG despite its efficiencies in item generation. Notably, these shortcomings relating to item simplicity and the cost of template design have been substantially mitigated by recent LLM-based approaches to AIG. For example, Chan et al. (2025) demonstrated that simply using prompt engineering, GPT-3.5 and GPT-4 could generate question-answer pairs with much flexibility across various technical domains (see the next section).

Finally, there are challenges in quality assurance. In theory, a single template can generate thousands of items, rendering comprehensive expert review impractical. Although Gierl and Lai (2016) propose iteratively reviewing small samples to manage this process, they

acknowledge that such methods remain labour-intensive, time-consuming, and cannot eliminate errors entirely. These persistent issues of quality control and evaluation are worth revisiting in more depth and considering in light of newer approaches to AIG that claim to reduce manual oversight while maintaining or even improving item quality.

2.2. A modern approach: LLMs for AIG

2.2.1. Hard prompting strategies for AIG

The rapid advancement of large language models (LLMs) is reshaping the possibilities for AIG. One particularly promising development lies in the use of prompt engineering, which draws on the LLMs' generative capacity without requiring changes to their underlying model parameters (Aryadoust et al., 2024). Specifically, hard prompt engineering involves crafting specific instructions or contextual cues to elicit desired output behaviours from LLMs, and has proven to be a lightweight, flexible, and cost-effective technique for automatic item generation (Chan et al., 2025). A comprehensive review by Chen et al. (2025) detailed the efficacy of various hard prompt engineering strategies. They argued that for many tasks, strategic prompt design significantly enhances model performance without modifying the underlying model weights, which positions prompt engineering as a resource-light mechanism for achieving improved and more aligned LLM behaviour. In the AIG domain, a growing body of work has applied prompt engineering to harness LLMs for item generation (Aryadoust et al., 2024; Attali et al., 2022; Biancini et al., 2024; Chan et al., 2025; Tran et al., 2023). The most basic approach involves supplying a short task description to guide the model's output. While this approach is attractive for its simplicity, several studies show that minimal prompts—often comprising of two or three instructional sentences—yield mixed outcomes. For example, Tran et al. (2023) found that models can generate items that resemble multiple-choice formats, whereas they also frequently mislabel distractors, sometimes even inserting correct answers into the distractor set. This underscores that while LLMs demonstrate promise in producing test-like items, they cannot yet do so consistently without oversight.

Attempts have been made to mitigate these issues through the use of more complex hard-prompting strategies. For example, Mucciaccia et al. (2025) and Scaria et al. (2024a) explored advanced techniques such as one-shot prompting, in which a single exemplar item is provided to guide generation; chain-of-thought prompting, which elicits step-by-step reasoning before finalising questions and options; and role-based prompting, where the model is assigned the persona of a trained test developer to encourage domain-appropriate item design. Scaria et al. (2024a) also varied the level of detail in different prompts. Both studies found that detailed instructions incorporating the aforementioned techniques outperformed simple prompts in generating coherent and higher-quality questions. In another study, Doughty et al. (2024) employed a similar strategy by supplying extensive guidelines that defined what constitutes a stem, key, and distractors, along with explicit rules and formatting requirements. With such scaffolding, LLMs were able to produce items closer in quality to human-authored ones. Despite these advances, even here, recurring problems emerged in the generated output, particularly with distractors that were implausible or inadvertently contained the correct answer—a limitation echoed in Yao et al. (2025). While Yao et al. (2025) attempted to overcome this through self-critiquing prompts, where the model iteratively evaluates and revises its own output, they found limited success. Performance often plateaued or deteriorated after a few cycles, highlighting that automated self-correction lacks robustness without external human input.

Finally, Hwang et al. (2024) and Shimmei et al. (2025) examined the effects of few-shot prompting on item generation. Few-shot prompting refers to the technique of providing an LLM with a small number of example inputs and outputs within the prompt to guide the generation of new items that follow similar patterns or structures. Hwang et al. (2024) and Shimmei et al. (2025) reported that supplying a small number of

worked example improved outcomes compared with zero-shot baselines, where no example input and outputs are provided within the prompt to the LLM. This suggests that providing exemplars enables the model to capture task structure more effectively, yielding measurable gains in performance. Nonetheless, Hwang et al. (2024) also identified a key limitation: even with examples included, the models frequently failed to meet more nuanced requirements, such as aligning items with specific Bloom levels. They concluded that such alignment errors point to the need for alternative approaches, such as fine-tuning, to better calibrate outputs to specific pedagogical objectives.

In sum, while hard prompting strategies can elicit test items from LLMs, they remain far from fool proof and still demand substantial human oversight and evaluation. Generated outputs frequently contain factual or structural errors, and even when useable, do not reliably align with more nuanced requirements. Thus, alongside efforts to overcome the performance plateau in hard-prompted items, a rigorous evaluation framework is crucial as they enable the systematic assessment, comparison, and refinement of item quality.

2.2.2. Evaluating automatically generated items at scale

Evaluation is a critical stage in the automatic item generation process as items intended for large-scale assessment must undergo systematic review of their psychometric properties to ensure validity, reliability, and fairness (Circi et al., 2023). However, current practices reveal several major shortcomings (Moon et al., 2022; Moore et al., 2023). Tan et al. (2025) noted that although some studies included human raters, few engaged subject matter experts or measurement specialists. As a result, the current literature provides limited evidence on whether automatically generated items meet the standards of quality required for educational use in teaching and assessment. One reason for this gap is that expert-based evaluation is highly resource-intensive. For example, Scaria et al. (2024a) reviewed over 2500 items, a process that required substantial time and effort from evaluators. Consequently, many studies that rely on human review adopt abbreviated rubrics with fewer than ten metrics (e.g., clarity, grammar, and answerability), or ask raters to score items on only a small set of dimensions such as complexity, correctness, or alignment with learning objectives (Mucciaccia et al., 2025; Scaria et al., 2024b; Shimmei et al., 2025).

In response to the limitations of human review, studies have begun experimenting with two types of automated methods: rule-based metrics and LLM-based evaluators. Rule-based approaches typically rely on detecting item-writing flaws, such as implausible distractors or ambiguous keys, and have been shown to perform at levels comparable to human judges in some cases (Moore et al., 2023). They also offer the advantage of interpretability and consistency, making them more statistically rigorous. By contrast, LLM-based evaluation has been tested on tasks such as assessing grammar, format, or relevance (Moore et al., 2023; Mucciaccia et al., 2025). These models show promise for simple, objective criteria, but their performance weakens on more subjective judgements such as option validity or alignment with core themes. Importantly, the prompts used for LLM evaluation have so far been minimal, which limits model performance and requires further research into prompt refinement for LLM evaluation (Moore et al., 2023). Compounding these challenges is the lack of shared standards—currently, no comprehensive theoretically grounded framework exists for systematically evaluating stems, keys, and distractors against best practices in MCQ construction (Grevisse, 2024). Evaluation rubrics and criteria vary widely, making it difficult to compare results across studies or to define what constitutes high-quality output.

This absence of a unified evaluative framework in educational assessment stands in contrast to the field of natural language processing (NLP), where widely recognised benchmarks provide common reference points that enable systematic comparison of LLM performance across tasks and domains. Some of the commonly used benchmarks include BLEU (Papineni et al., 2002), a bilingual evaluation understudy for measuring n-gram overlap; ROUGE (Lin, 2004), a recall-oriented

understudy for gisting evaluation for assessing summaries; METEOR (Banerjee and Lavie, 2005), a metric for translation quality that accounts for synonymy and stemming; and MMLU (Hendrycks et al., 2021), a massive multitask language understanding benchmark for evaluating general knowledge and reasoning. By contrast, in AIG research, many studies adopt either ad hoc criteria or repurpose NLP metrics in ways that are only partially suited to assessment contexts. For example, Moore et al. (2023) note that while readability or explainability scores based on BLEU or METEOR are sometimes used, these measures correlate poorly with human evaluations and fail to account for the pedagogical value of questions. Other authors have turned to customised rubrics: Scaria et al. (2024a) used a nine-item checklist covering clarity, grammar, and understandability, while Mucciaccia et al. (2025) employed five criteria including format, language accuracy, and option validity. While valuable within their respective studies, such bespoke rubrics hinder cross-study comparability and make results dependent on subjective definitions. Reflecting on these limitations, Kurdi et al. (2020) explicitly call for harmonised evaluation metrics. Without a coherent and broadly applicable evaluation set of standards, the field could struggle to advance cumulative knowledge.

In sum, two central gaps emerge from the literature reviewed thus far. First, hard prompt engineering, while lightweight and flexible, eventually plateaus in its ability to consistently produce accurate items and to meet more nuanced item requirements. Second, there is no standardised evaluation framework for AIG. Together, these issues underscore that the field struggles not only with generation but also with systematically verifying quality at scale.

2.3. Academic listening subskills and test item design

2.3.1. The construct of academic listening

Academic listening is widely understood as a multidimensional construct involving the coordinated deployment of multiple interacting subskills rather than a single, unitary ability (Aryadoust, 2020; Aryadoust & Luo, 2023; Buck, 2001; Eom, 2008, pp. 77–94; Field, 2008; Liao, 2009). Within the lecture comprehension context that forms the focus of the present study, these subskills can be understood as falling along a continuum from lower-order, locally-focused processing to higher-order, globally-oriented comprehension. Drawing on established taxonomies in L2 listening research (Aryadoust, 2017; Aryadoust & Goh, 2013; Aryadoust & Luo, 2023; Buck, 2001; Field, 2008; Liu & Aryadoust, 2026; Liu et al., 2022; Shin et al., 2021), the present study operationalises academic listening ability through eight discrete but interrelated subskill categories, each of which targets a distinct cognitive demand placed on the listener.

At the local comprehension level, **Detail** items require listeners to identify specific, explicitly stated information in the lecture, such as facts, figures, or events (Aryadoust, 2017, 2020; Liu et al., 2022; Shin et al., 2021). **Gist-Content** items, by contrast, target global comprehension of the main topic or central idea of the lecture as a whole, requiring the listener to synthesise information across extended stretches of discourse (Aryadoust, 2017, 2020). **Gist-Purpose** items similarly address global comprehension but focus specifically on the communicative or rhetorical purpose of the lecture (In'nami & Koizumi, 2012).

At the inferential level, **Inference** items require listeners to derive meaning that is implied but not explicitly stated (Aryadoust & Goh, 2013; Eom, 2008, pp. 77–94; Field, 2008; Liao, 2009; Liu et al., 2022; Shin et al., 2021). **Attitude** items target a specific form of inferencing focused on the speaker's stance, feelings, or evaluative orientation toward the content being discussed, which requires sensitivity to prosodic and pragmatic cues embedded in the discourse (Aryadoust, 2017). **Function** items assess the listener's ability to interpret the communicative function of specific utterances or exchanges, which often requires distinguishing between what is said and what is meant (Aryadoust & Luo, 2023).

At the discourse level, **Organisation** items evaluate comprehension of how information is structured within the lecture. Finally, **Connecting Content** items require listeners to integrate information across disparate sections of the lecture or across multiple speakers in a discussion format, making connections between ideas that are not explicitly linked in the discourse. These higher-order subskills place great demands on working memory and discourse-level processing, and are consequently the most challenging to elicit reliably through test items (Buck, 2001; Field, 2013).

In sum, these eight subskill categories discussed above reflect the hierarchical and interactive nature of academic listening comprehension, spanning from surface-level detail recognition to sophisticated discourse-level integration. This taxonomy provides the theoretical basis for the item generation and evaluation framework developed in the present study.

2.3.2. Operationalising academic listening subskills through MCQ design

Translating the listening construct into valid and reliable test items is a non-trivial challenge. One of the most widely used approaches for operationalising listening constructs is the use of multiple-choice questions (MCQs), which are prevalent in academic listening assessment and must address several competing demands (Buck, 2001; Field, 2013; Holzknecht et al., 2021; O'Grady, 2021). At the most fundamental level, items must be grounded in the passage; in other words, they must require access to the listened content rather than being answerable through general world knowledge or test-wise strategies alone (Buck, 2001; O'Grady, 2024). This passage-dependence is a prerequisite for construct validity: items that can be answered without reference to the lecture provide no interpretable evidence of listening ability (Field, 2013).

A key challenge in MCQ design for listening assessment concerns the relationship between item stems and the linguistic content of the passage. Stems that paraphrase or closely mirror the wording of the passage risk cueing the correct answer through lexical matching rather than genuine comprehension, thereby rewarding recognition rather than listening. Conversely, stems that are too distant from the passage may introduce construct-irrelevant difficulty, penalising listeners not for their comprehension but for their inability to navigate the item's linguistic demands. Principled MCQ design therefore requires stems that are sufficiently paraphrased to prevent cueing while remaining unambiguously linked to specific, identifiable passage content (Haladyna, 2022; O'Grady, 2021).

The design of distractors in MCQs presents an equally important challenge. Effective distractors must be plausible; in other words, they must be grounded in the passage in a way that makes them superficially attractive to listeners who have not fully comprehended the relevant section (Aryadoust, 2017), while remaining clearly incorrect upon careful listening (Buck, 2001; Haladyna, 2022). Implausible distractors that are easily eliminated reduce the diagnostic value of items by making correct responses accessible through elimination rather than comprehension. At the same time, distractors that are overly similar to the key (the correct option), or that introduce alternative correct interpretations, undermine test item validity by rendering the answer ambiguous (O'Grady, 2021). In addition, according to Haladyna (2022), the avoidance of structural cues, including longest-correct-option bias, absolute terms, and grammatical inconsistency between stems and options, is a further requirement of principled MCQ design, which is particularly relevant to the automated generation context examined in this study.

Different listening subskills, discussed in previous section, place distinct demands on item design. Specifically, detail items can typically be written by identifying a specific, unambiguous information unit in the passage and constructing distractors from related but incorrect information in the same vicinity (Aryadoust, 2017; Buck, 2001). Inference and Attitude items, by contrast, require distractors that are plausible from a surface reading but inconsistent with a deeper interpretation of

the speaker's intent, which places greater demands on the item writer's ability to anticipate likely misinterpretations (Field, 2013). Organisation and Connecting Content items are particularly difficult to construct reliably, as they require both stems and distractors to reflect meaningful distinctions in discourse structure that are often subtle and context-dependent (Aryadoust & Luo, 2023; Field, 2013). These differential demands across subskill types are directly relevant to the evaluation findings reported in the present study and inform the interpretation of results across item categories.

2.4. Fine-tuning LLMs for automated item generation (AIG)

Whereas hard prompting steers model behaviour externally, shaping outputs without altering the underlying parameters, fine-tuning directly updates the model's parameters through additional training on a curated dataset (Zhang et al., 2025). This process reduces reliance on ad hoc prompt design, improves alignment with task requirements, and allows the model to internalise domain-specific patterns, response formats, and task structures that prompt engineering alone might not reliably enforce (OpenAI, n.d.-b). Fine-tuning can therefore be positioned as a natural next step once the gains from prompt engineering have plateaued, offering a way to move beyond the performance limits of prompt-only methods (OpenAI, n.d.-b). In AIG, the training dataset consists of authentic test samples, represented as annotated input-output pairs where each input prompt specifies task instructions and contextual information (e.g., required schema and content) and each output is an authentic test item. This setup enables the model to learn not only what to generate but also how to format its responses in line with assessment conventions. As a result, fine-tuning enhances task fidelity, particularly in domains requiring structured outputs and specialised knowledge, such as language assessment and education (Wang et al., 2025).

As outlined above, there is a common assumption that fine-tuning on a sufficiently large set of well-structured training examples—in this case, listening test samples—has the potential to yield high-quality outputs (Chung et al., 2022). However, relatively little work has examined the effectiveness of fine-tuning specifically for open-ended task generation tasks such as language assessment tasks, with most work to date focusing instead on predictive and constrained tasks such as classification, translation tasks, and single-question generation (Bucher & Martini, 2024; Sivakaran & Yang, 2025; Yang et al., 2024). This gap is significant as the challenges of generating complex, structured assessment items differ substantially from those of more constrained, predictive tasks. In tightly defined applications such as classification or translation, task objectives and output formats are narrow and well specified, which means that model behaviour is naturally bounded and prompt design plays a comparatively minor role. By contrast, in open-ended tasks such as generating listening test items, the instruction prompt carries far greater weight: it must guide the model not only toward linguistic correctness but also toward balancing creative flexibility with adherence to pedagogical and structural requirements (Rae et al., 2022). Poorly designed prompts risk introducing misleading training signals that diminish the effectiveness of fine-tuning, even when the dataset itself is otherwise of high quality. For this reason, prompt engineering and fine-tuning should not be seen as isolated strategies but as potentially interacting processes in model adaptation. Accordingly, this study treats prompt engineering and fine-tuning as complementary methods, with prompt design guiding both inference-time performance and the fine-tuning process itself.

2.5. The present study

Our experimental work has two main goals. First, it systematically investigates iterative prompt engineering as a structured optimisation process, examining how successive refinements influence item quality under a prompt-only regime. Rather than treating prompting as a static baseline, the study analyses how targeted modifications to instruction

design shape model behaviour and identifies the point at which performance plateaus. Second, it explores supervised fine-tuning as a means of moving beyond the constraints observed in prompt-only approaches, evaluating whether model-level adaptation yields more stable and construct-aligned outputs.

Second, the study proposes a systematic and scalable two-tiered evaluation framework grounded in established test development principles to assess item quality across conditions. Building on the gains and limitations identified through iterative prompt refinement, fine-tuning is examined as a subsequent stage of optimisation, offering a mechanism to align models more directly with domain-specific assessment requirements. Specifically, the present study investigates two central research questions:

1. To what extent can iterative prompt engineering improve the quality of generated items for language assessment, and where does its performance plateau?
2. What is the marginal benefit, if any, of fine-tuning over a prompt-only approach?

3. Method

3.1. Phase 1: zero-shot prompt engineering with base GPT-4.1 model

The objective of the first phase was to analyse the effect of prompt quality alone, independent of any fine-tuning or exposure to training examples, on the generation behaviour of the base GPT-4.1 model, using OpenAI's API. Specifically, this phase adopted a zero-shot approach in which no task-specific examples were provided to the model during inference. Instead, model outputs were shaped solely by the quality and structure of the instruction prompt. To this end, we employed an iterative prompt refinement strategy, illustrated in Fig. 1, comprising of three iterations (Table 1). Each iteration was built upon a consistent system message (Supplementary Material 1) that established the model's role and task scope.

Building on this foundation, the first iteration introduced an initial instruction prompt (Supplementary Material 2.1) designed to provide clear task framing, content constraints, and output formatting guidance (OpenAI, n.d.-c). Importantly, the prompt specified the expected output structure and enforced formatting consistency through explicit section labels and style requirements (White et al., 2025). In designing the prompt, we adopted a subskill-based approach to listening assessment (Aryadoust & Luo, 2023; Buck, 2001; Field, 2008). The construct of academic listening was operationalised through discrete but interrelated comprehension subskills, including gist understanding, detail recognition, inference, speaker attitude, and discourse organisation. These subskills were instantiated through predefined question tags (Supplementary Material 3), ensuring that generated items were anchored to specific listening subskills rather than to generic comprehension tasks. Furthermore, the prompt incorporated established multiple-choice item-writing principles (Haladyna, 2022), including guidelines on distractor plausibility, option independence, avoidance of cueing, and clarity of stems. This ensured that prompt refinement was grounded not merely in surface-level formatting compliance but in professional standards of principled assessment design.

Notably, the prompt did not explicitly encode a formal proficiency framework (e.g., CEFR bands) or assessment purpose (e.g., formative

Table 1

Prompt iterations: design choices.

Iteration	Design
1	• Role-prompting system message ("You are an AI assistant that generates listening tests"), improving controllability (Chen et al., 2025). • Adopted basic prompt-engineering principles: clear, specific, precise wording (Chen et al., 2025); baseline template (lecture, questions, tags, answers). • No explicit rules or constraints.
2	• Introduced detailed MCQ writing guidelines (Haladyna, 2022). • Added contextualisation for richer task framing, consistent with alignment findings (Radford et al., 2019). • Made constraints more declarative/visible through a Rules section (Reynolds and McDonnell, 2021).
3	• Refined flow/structure of instructions for clarity and concision. • Rephrased rules to directly target key issues (answer-key distribution, longest-correct-option bias). • Incorporated advanced strategies, incl. zero-shot chain-of-thought prompting for stepwise option drafting (Sahoo et al., 2025), to even key distribution and standardise option length. • Converted soft guidelines into measurable rules/thresholds, consistent with operationalised constraints (Liu et al., 2025).

versus summative use). However, the absence of explicit proficiency labelling in the prompt design does not mean that the generated items are proficiency-agnostic. The fine-tuning dataset was drawn from sample test materials of TOEFL-aligned iBT listening, and although these sample test materials are not authentic TOEFL tests, they closely resemble and can reasonably be considered representative of the listening construct assessed by the TOEFL iBT Listening section. The TOEFL iBT is explicitly designed and validated to differentiate test taker ability across the full range of CEFR proficiency levels, from A1 upward, using a multistage adaptive methodology that matches task difficulty with learner proficiency (Manna et al., 2025). Specifically, Manna et al. (2025, p. 2) mentioned that, "[t]he TOEFL test is intended for older adolescents and adults who wish to provide evidence of their overall English language proficiency level in academic and daily life settings." In addition, "In terms of proficiency levels described in the Common European Framework of Reference (CEFR; Council of Europe, 2001; 2020), the TOEFL test is designed to cover the full range from A1 to C2" (Manna et al., 2025, p. 2).

The foundational skills dimension of the TOEFL construct is specifically intended to support differentiation across this full proficiency spectrum (Manna et al., 2025). By training on sample test items drawn from this assessment, the fine-tuned model implicitly inherits the proficiency-sensitive characteristics of the training data, even without explicit CEFR labelling at inference time. Assessment purpose and proficiency level were therefore held constant and implicitly defined through the use of this sample of academic lecture discourse and university-style comprehension tasks, which are characteristic of upper-intermediate to advanced academic listening settings. The generated test items were intended to reflect this fixed academic listening construct rather than to model variation across proficiency levels or assessment purposes. Accordingly, the study evaluates how prompt refinement and fine-tuning influence item quality within this defined context, rather than investigating alignment across a broader range of external proficiency standards. Formal scale calibration and proficiency-level validation across the full CEFR range fall beyond the

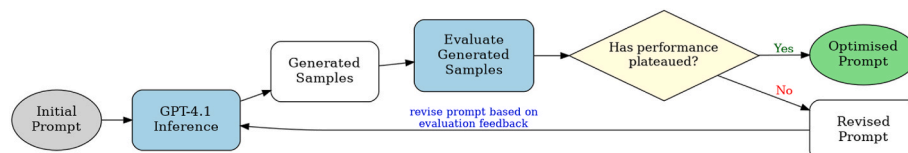


Fig. 1. Iterative prompt refinement strategy across three phases.

scope of the present experimental design and are identified as important directions for future work.

Next, the prompt was submitted to the base GPT-4.1 model via the OpenAI API, and ten independent tests (where each test consisted of a passage and six items) were generated through separate inference calls. The generated tests were then evaluated on a set of metrics, detailed in Section 3.4.2. At the end of the first iteration, we drew on evaluation findings and established prompt-engineering principles (Chen et al., 2025) to revise and improve the instruction prompt. This revised version served as the starting point for the second iteration, which was again submitted to the model to generate ten new tests for evaluation. The process formed an iterative feedback loop (Fig. 1) in which each cycle of generation and evaluation informed further refinements. Iterations continued until performance gains began to plateau, indicating diminishing returns in output quality. The final instruction prompt at this stage was taken to represent the optimised version (iteration 3). The mechanisms underpinning each iteration are summarised in Table 1. This phase functions as a prompt ablation study, which allows for systematic observation of how changes in prompt design impact the model's output quality and task alignment abilities.

3.2. Phase 2: fine-tuning GPT-4.1

The second phase of the study investigated the effects of fine-tuning on the quality and consistency of generated listening items. As previously outlined, fine-tuning involves updating model parameters using training data in the form of input-output pairs (instruction prompts paired with high-quality listening tests). The goal was to extend the capabilities of prompt-only methods by moving beyond surface-level control of outputs. To this end, we fine-tuned the model using the optimised instruction prompt from Phase 1 (Supplementary Material 2.3), with the objective of enabling the model to internalise task-specific structures and aligning its performance more closely with the desired outputs.

3.2.1. Fine-tuning data preparation

As fine-tuning is known to be highly sensitive to the quality of training data, especially in specialised domains (Sun et al., 2024), particular attention was paid to the content fidelity and structural consistency of the fine-tuning dataset. The dataset consisted of 42 listening tests drawn from publicly available TOEFL-style test preparation materials. One of the authors reviewed these tests to evaluate their content and the construct they were designed to measure. Each test included the full lecture transcript, corresponding multiple-choice questions, and the answer key. The data was cleaned to enforce consistent formatting and structure across the entire dataset. After data cleaning, feature engineering was performed to augment each test with structured metadata. This served to enhance the model's capacity to produce controllable and well-aligned outputs. The engineered features included:

- Question type tags (Supplementary Material 3), to specify what listening subskill the test item likely measured (see Aryadoust & Luo, 2023);
- Topic tags, chosen from a predefined list of content domains (Aryadoust & Luo, 2023; Supplementary Material 4);
- Lecture type, indicating whether the listening passage is a lecture with classroom discussion featuring student-professor interaction and turn-taking, or a lecture without discussion, delivered as a monologue by the professor; and
- Question counts, specifying the number of items per test.

Detailed corpus-level metadata statistics can be found in Supplementary Material 5. This metadata was dynamically injected into the prompt template and dataset at fine-tuning, and served several key functions. First, the topic and question count fields allowed for

controlled, fine-grained prompting at inference-time (Fan et al., 2025), enabling test developers to specify both the desired subject matter and number of questions to be generated per test. This helped to reduce the variability typically associated with generative models, as the model learns to associate specific metadata inputs with structured, predictable outputs. Second, the lecture type distinction helped the model learn the pragmatic and structural differences (Beelen & van Strien, 2022) between discourse types, such as shifts in speaker roles, use of discourse markers, and turn-taking conventions. Finally, the inclusion of question type tags representing the listening subskills elicited by each question (Aryadoust & Luo, 2023) facilitated subsequent quantitative evaluation of the model's outputs. By training the model to associate specific question types to their corresponding categories and then instructing the model to label generated questions with categories in its output, we could analyse the distribution of question types in generated tests. This allowed us to quantitatively assess how well the model internalised and reproduced authentic question distributions in its outputs.

The procedures outlined in this section facilitate the development of a fine-tuning dataset that supports higher-fidelity learning and enables evaluation grounded in statistical analysis. Furthermore, the fine-tuning dataset was curated to reflect theoretically meaningful variation in lecture type, topic, and question function, thereby reinforcing the task model's alignment with the intended listening construct rather than optimising surface-level fluency alone. This dataset was used in tandem with the optimised prompt from Phase 1 to fine-tune GPT-4.1, as discussed next.

3.2.2. Fine-tuning GPT-4.1

The fine-tuning was conducted on OpenAI's API using the following hyperparameters: three epochs, a batch size of two, and a learning rate multiplier (LRM) of 0.05. In this context, hyperparameters are configuration settings specified prior to training that govern how learning occurs. The number of epochs defines how many times the model iterates over the training dataset; increasing this value may improve learning but also raises the risk of overfitting—that is, when a model captures noise or idiosyncrasies in the training data and consequently performs poorly on unseen examples. Batch size refers to the number of training examples processed together before the model's parameters are updated; smaller batch sizes, such as two, are often used to reduce memory usage and can stabilise training updates (Goodfellow et al., 2016). The LRM adjusts the base learning rate to control how quickly the model adapts; a lower LRM (e.g., 0.05) ensures more gradual updates, which is particularly appropriate when fine-tuning LLMs to avoid catastrophic forgetting—when a model learns new information and, in the process, forgets what it had previously learned (Kirkpatrick et al., 2017)—or overshooting, where updates exceed the optimal point during training and cause instability rather than improvement (Goodfellow et al., 2016). These values were selected based on general best practices outlined in OpenAI's fine-tuning documentation (OpenAI, n.d.-a) and similar guidelines from prior studies on instruction tuning (Chung et al., 2022; Zhang et al., 2025). Given the focused nature of our dataset and the instruction-driven nature of the task, these settings provided a conservative yet effective balance between learning stability and adaptation.

After each fine-tuning run, the model was used to generate 10 new tests using the same instruction prompt used during training, along with specified topics (Supplementary Material 4). These generated tests were later evaluated on the metrics described in Section 3.4.2. These experiments mirror Phase 1 but assess the effect of prompt quality when used in the training loop. They provide a direct test of whether suboptimal prompt design during training can limit or distort model learning, even when data examples are high-quality. Following each fine-tuning run, the model was prompted ten times to generate a distinct set of outputs for evaluation. Each prompt included the same instruction used during training, ensuring alignment between training and inference conditions. These experiments parallel those in Phase 1 but shift the focus to the role

of prompt quality within the learning process itself.

3.3. Phase 3: isolating the marginal benefit of fine-tuning GPT-4.1

3.3.1. Zero-shot prompting vs. fine-tuning on GPT-4.1

The final phase of this study aims to isolate the contribution of fine-tuning to model performance when prompt quality is held constant. Specifically, it evaluates whether fine-tuning offers meaningful gains over prompt engineering alone in the context of listening assessment item generation. We compared the results of two conditions: iteration 3 of Phase 1, which used the optimised instruction prompt to run zero-shot inference on the base GPT-4.1 model, and Phase 2, which used the same prompt—augmented with metadata (Supplementary Material 4)—for fine-tuning on the full 42-test dataset. By holding the prompt design constant across both experiments, this phase examines the isolated impact of model adaptation through fine-tuning, independent of prompt variation. This comparison is particularly relevant for L2 test developers operating under resource constraints. If prompt engineering alone yields comparable performance to fine-tuning, it may serve as a more cost-effective and operationally efficient alternative in many testing

scenarios.

3.4. Evaluation of generated items

As discussed in Section 2.2.2, evaluation is a central challenge in AIG, and prior work shows that neither human nor automated methods alone are sufficient to ensure validity, reliability, and scalability. To address this gap, we propose an eclectic two-tiered evaluation framework that integrates deterministic (rule-based) methods and (minimised) human review. The design principle is to assign each evaluation task to the method best suited for it, thereby combining statistical rigour, coverage of complex dimensions, and practical feasibility at scale.

3.4.1. Rationale for a two-tiered evaluation framework

The objective of this hybrid evaluation framework is to first maximise the use of deterministic, rule-based methods, then selectively allow for subjective metrics to minimise human reliance. Rule-based methods are prioritised wherever possible because they are deterministic, interpretable, and statistically rigorous. Notably, unlike human-based

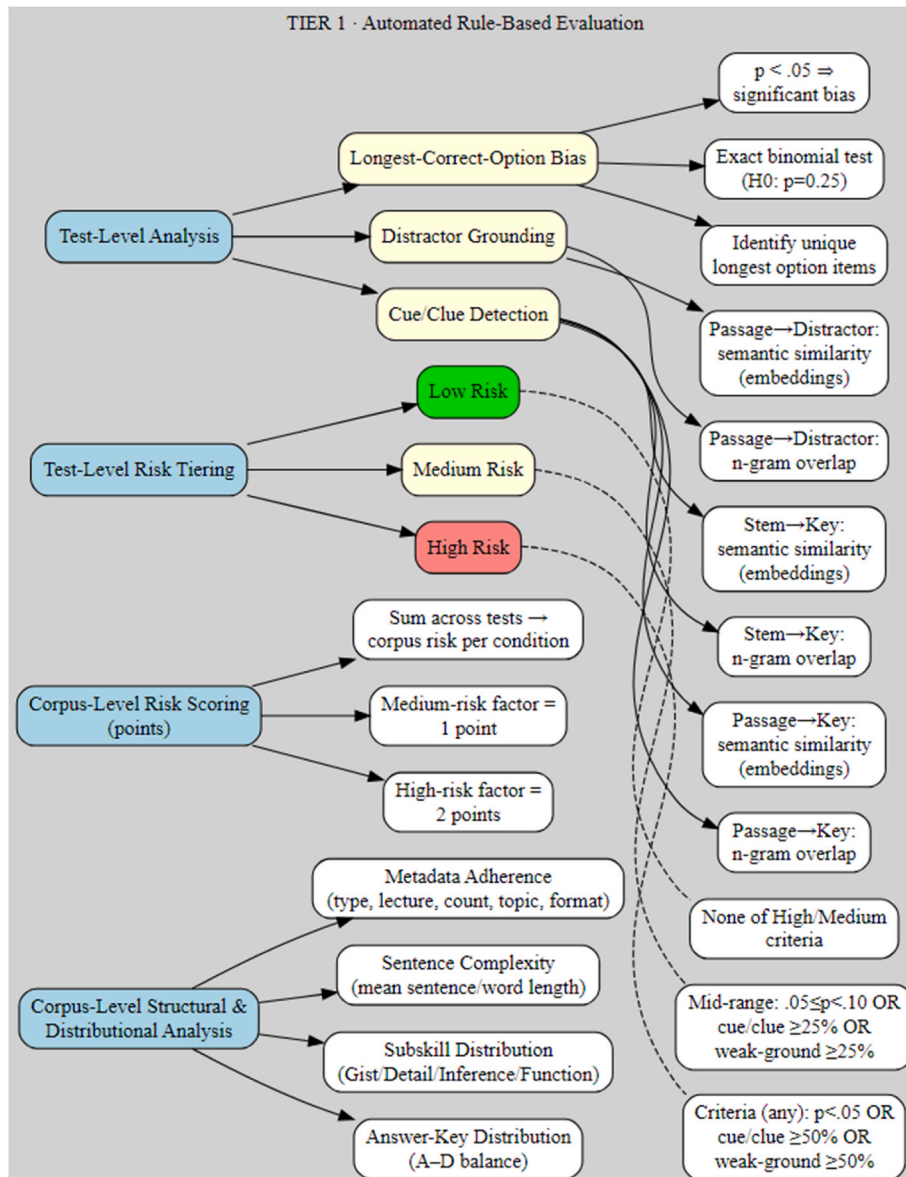


Fig. 2. Operationalisation of Tier 1 or the automated rule-based evaluation in the two-tiered evaluation framework.

evaluation, their outcomes can be replicated without variation. Human-based evaluation is introduced where rule-based methods cannot capture more subjective or nuanced constructs. For example, assessing option plausibility or identifying whether items are “trick questions” requires context-sensitive reasoning that deterministic metrics cannot approximate. Delegating the first cut of evaluation to rule-based methods allows for minimisation of human evaluation. This is a deliberate decision that allows for reduction of cost, labour-intensiveness, and reliance on specialised expertise.

In AIG research, quality assurance is not a concern separate from validity but rather its operational instantiation at the item level (Alhazmi et al., 2024; Circi et al., 2023). Systematic checks on distractor plausibility, passage grounding, and structural independence are the mechanisms through which potential construct-irrelevant factors can be identified and minimised. The present framework should therefore be understood as a validity-oriented evaluation instrument grounded in established AIG and assessment principles, rather than a checklist of surface-level formatting requirements.

3.4.2. Two-tiered evaluation framework

The two-tiered evaluation framework was structured around key item-quality dimensions derived from established test development principles (Haladyna, 2022). Each dimension was operationalised through a hybrid method that combined deterministic rule-based checks and human review as summarised in Figs. 2 and 3 and described in detail in subsequent sections. The framework operates as a *tiered* pipeline, as follows:

- Tier 1 (Fig. 2): Automated rule-based checks were applied first to all items, identifying basic structural or statistical flaws. This evaluation tier was applied to systematically screen and flag structural and statistical weaknesses in the model-generated items at scale. It was not used to replace human judgment, but to provide reproducible diagnostics that highlighted potential flaws for further inspection.
- Tier 2 (Fig. 3): Human expert evaluation was then performed on all items, focusing on complex dimensions requiring interpretive judgement that could not be covered in Tier 1.

This integrated framework ensured that evaluation was both scalable and scientifically grounded. Ten complete listening tests were generated for each experimental condition (prompting iterations 1, 2, and 3, and the fine-tuned model), yielding a total of 40 tests and 240 multiple-choice items (each test contained a lecture transcript and six items with answer keys).

3.4.3. Tier 1: rule-based evaluation

After parsing the 10 complete test outputs from each of the four experimental conditions (40 tests; 240 items in total), per-item and per-test metrics were accumulated based on predefined criteria and fixed thresholds.

Test-level analysis. These metrics were aggregated over each test to produce summary statistics and test-level risk assessments:

1. Answer key cue/clue detection:
 - o. Passage-to-key lexical and semantic overlap. N-gram overlap and semantic similarity via sentence embeddings were computed between the lecture (or its sentences) and the correct option.
 - o. Stem-to-key lexical and semantic overlap. Similarly, overlap and semantic similarity between the question stem and the correct option were evaluated.

For each of the above dimensions, tests were flagged if the proportion of items where both N-gram and semantic similarity metrics surpassed a specified cut-off value (Supplementary Material 6), thereby ensuring that options were not flagged for normal paraphrasing but only when they were excessively similar in wording to the passage or stem.

2. Distractor grounding: Each distractor was examined for how grounded it was in the passage. If either its N-gram overlap or semantic similarity with the lecture was below a threshold (Supplementary Material 6), the distractor was flagged as having weak passage grounding, making it more likely to be an implausible distractor. The proportion of items in the test containing at least one flagged weak distractor was computed.
3. Binomial test for longest-correct-option bias: To evaluate whether the correct option tended to be the longest more often than expected by chance, each item with a unique longest option was treated as a trial. Under the null hypothesis, the probability of the correct answer being longest is 0.25, reflecting the one-in-four chance associated with four options. The observed number of “longest-correct” cases within a test was compared to this expectation using an exact binomial test. Tests yielding a p -value below 0.05 were interpreted as exhibiting statistically significant longest-correct bias.

Risk tiering. A composite risk tier was assigned to each test (High, Medium, Low) by combining the three test-level signals: cue/clue rates, weak distractor grounding, and longest-correct-option bias, which are linked to test-wiseness and item fairness:

1. A test was classified as High risk if any one of the following applied:
 - o. The binomial test for longest-correct-option bias returned $p < 0.05$.
 - o. Cue/clue rate in the test was $\geq 50\%$.
 - o. Weak distractor grounding rate in the test was $\geq 50\%$.
2. A test was classified as Medium risk if none of the High risk criteria were met, but at least one metric was mid-range ($0.05 \leq p < 0.10$ for the binomial test, cue/clue rate $\geq 25\%$, or weak distractor grounding rate $\geq 25\%$).
3. A test was classified as Low risk if it met none of the High or Medium risk criteria.

In addition to categorical tiering, a simple point-based scheme was also applied to quantify cumulative risk across conditions. For every item, the presence of one factor (i.e. significant longest-correct bias, weak distractor rate above threshold, or cue/clue rate above threshold) in the High or Medium risk range equated to two and one point(s), respectively, such that a single item could accumulate multiple points if more than one flaw was present. These points were then summed across all tests within each experimental condition, providing a corpus-level indicator of overall risk. Tracking this aggregate score across iterations allowed us to examine whether total risk decreased as prompts were refined and fine-tuning was introduced.

The automated Tier 1 evaluation provided diagnostic feedback across conditions and tests, identifying which experimental conditions showed systematic risk. Furthermore, it enabled the comparison of conditions (prompt iterations and fine-tuning) on structural and statistical dimensions, independently of human judgment.

Corpus-level analysis. In addition to test-level checks, we examined global distributions across the full corpus of generated items to capture systematic trends and compare them against ground truth distributions from our dataset of authentic listening tests. Four dimensions were considered:

1. Answer key distribution: the overall frequency of correct answers across options A-D, used to identify imbalances that may create test-wiseness effects.
2. Question type distribution or subskill-targeting: the proportion of items tagged under each functional category (e.g., Gist, Detail, Inference, Function, etc. based on Aryadoust, 2017; Aryadoust & Luo, 2023; Field, 2008), which indicated whether the generated item sets reflected a balanced coverage of skills.
3. Sentence complexity: basic linguistic measures, including average sentence length in words and average word length, which provided

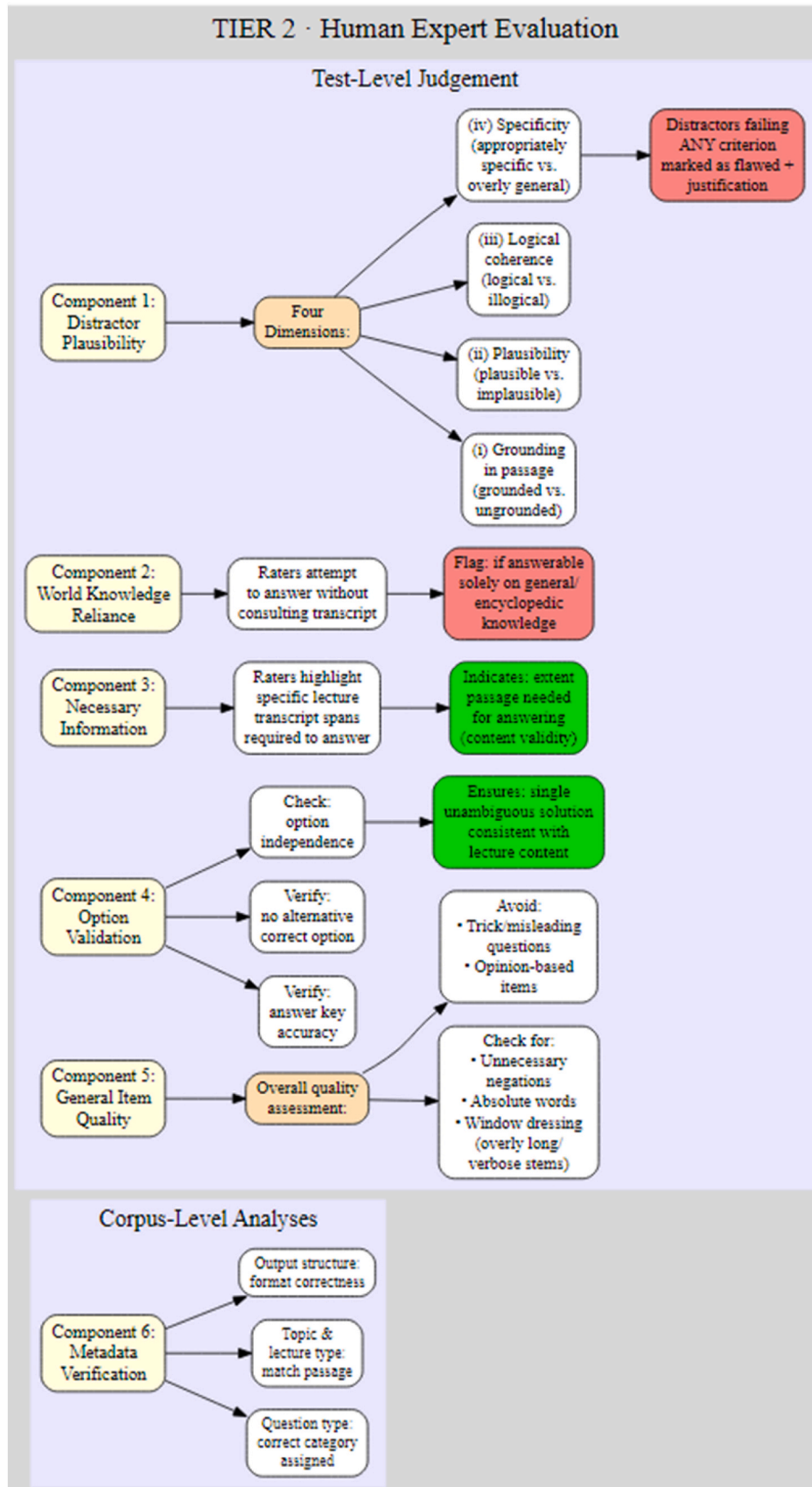


Fig. 3. Operationalisation of Tier 2 or the automated rule-based evaluation in the two-tiered evaluation framework.

an indication of linguistic variety and readability across stems and options.

4. Adherence to metadata requirements:
 - o. Correct question category assigned to each item.
 - o. Lecture type (monologue or classroom discussion), number of items per test, and lecture topic are aligned with the metadata requirements.
 - o. Correct overall output format and structure.

3.4.4. Tier 2: human evaluation

Tier 2 incorporates expert human review to assess interpretive dimensions of item quality that could not be reliably captured by automated rule-based methods. Raters used a structured annotation interface, informed by previous research in L2 listening assessment (Aryadoust et al., 2024; Buck, 2001; O'Grady, 2024) to evaluate each question against a standardised set of categories: distractor plausibility, reliance on world knowledge, necessary information, option validation, and general item quality.

1. Distractor plausibility: each distractor was rated along four dimensions: (i) grounding in the passage (grounded vs. ungrounded), (ii) plausibility (plausible vs. implausible), (iii) logical coherence (logical vs. illogical), and (iv) specificity (appropriately specific vs. overly general). Distractors failing any of these criteria were marked as flawed, with raters providing a short justification in such cases. This ensured transparency and facilitated subsequent error analysis.
2. World knowledge reliance: to assess whether items could be answered without reference to the passage, raters attempted to answer each question without consulting the transcript. If a question could be correctly answered solely on the basis of general or encyclopaedic knowledge, it was flagged as over-reliant on world knowledge rather than listening comprehension.
3. Necessary information: raters highlighted the specific spans of the lecture transcript required to answer each item. This indicated the extent to which the passage was necessary for answering the question and provided a direct check of content validity.
4. Option validation: raters verified that the designated answer key was accurate and that no alternative option could also be considered correct. This ensured that each item had a single unambiguous solution consistent with the lecture content. Raters also considered whether options were independent of one another.
5. General quality: raters judged the overall quality of each item. This included checks for unnecessary negations, absolute words, window dressing (overly long or verbose stems), trick or misleading questions, and opinion-based items.
6. Corpus-level analyses: raters verified that the generated metadata was accurate. This included ensuring the question type was of the correct category, the topic and lecture type matched that of the passage, and the overall output structure was correct.

The human evaluation rubric is provided in [Supplementary Material](#)

7. Each dimension was rated using binary judgement, with explicit operational definitions to minimise ambiguity. Prior to formal evaluation, a norming session was conducted in which raters jointly reviewed the rubric and evaluated a small pilot set of items to clarify interpretation of each category. Disagreements were discussed to ensure consistent application of the criteria before proceeding to independent rating. Inter-rater reliability statistics and the content validity of the evaluation framework are reported in Section 3.5.

To provide an interpretive benchmark against which the quality of AI-generated items could be assessed, the Tier 2 human evaluation rubric was also applied to the fine-tuning dataset; that is, the TOEFL-aligned human-generated materials used to train the fine-tuned model. This dataset consists of professionally developed listening test items that have undergone our review, and therefore represents a meaningful reference point for professional assessment standards. A total of 245

items were evaluated across the same Tier 2 dimensions used to assess the AI-generated items. It should be noted that this evaluation was conducted by a different rater from those used in the primary evaluation. The results are nevertheless informative as an indicative benchmark and are presented alongside the AI-generated item results in [Table 5](#).

3.5. Reliability and validity of the evaluation framework

3.5.1. Inter-rater reliability

Human evaluation was conducted by two raters, both trained in test evaluation. Rater A is a research assistant in the field of L2 listening and language assessment, while Rater B is a PhD student in language assessment. Rater A evaluated all generated items, while Rater B independently evaluated a 20% subset (two tests from each experimental condition) to establish inter-rater reliability. The agreement rate between Raters A and B for the overlapping subset was 79%. Given the highly imbalanced distribution of several binary rating categories (e.g., rare flags for trick questions or negation), Cohen's kappa was unstable and yielded paradoxically low values despite high observed agreement — a well-documented statistical artefact under skewed marginal distributions (Gwet, 2008). Gwet's AC1 was therefore adopted as the more appropriate chance-corrected coefficient. Across categorical dimensions, mean AC1 values ranged from 0.60 to 0.70, indicating moderate-to-substantial agreement (Landis & Koch, 1977). These results suggest that human evaluation judgments were applied with acceptable consistency. Discrepancies were resolved through discussion, and all results reported in this study are based on the agreed-upon set of evaluations.

3.5.2. Validity of the evaluation framework

In the present study, evidence-centered design (ECD) served as the organising framework for linking the evaluation dimensions directly to construct validity. ECD treats assessment design as an evidentiary argument in which claims about learners, the evidence needed to support those claims, and the tasks designed to elicit that evidence must be explicitly aligned (Mislevy et al., 2003). Accordingly, the target construct was operationalised as academic listening ability, including comprehension of gist, detail, inference, speaker attitude, discourse organization, and connections among ideas, consistent with established accounts of listening assessment and academic listening (see [Section 2.3.1](#) for a full account; Buck, 2001; Field, 2008, 2012, 2013; Aryadoust, 2017; Holzknicht et al., 2021; O'Grady, 2021, 2024; Aryadoust & Luo, 2023). These construct claims were instantiated through task models in which GPT-generated lectures and MCQs were designed to elicit the targeted listening subskills through constrained question tags. The evidence model was then operationalised through a hybrid quality-control framework that evaluated whether the generated items were passage-dependent, appropriately paraphrased, supported by plausible but clearly incorrect distractors, and free from cueing or other construct-irrelevant features that could weaken the interpretation of item responses as evidence of academic listening ability (see [Table 2](#)).

Within this framework, quality control in AIG was treated as item-level validity evaluation. Our approach is consistent with the logic of AIG, where scalable item production must still be governed by principled item models, construct specifications, and validation checks (Buck, 2001; Gierl & Lai, 2012, 2013, 2016; Haladyna, 2022). The evaluation dimensions in [Table 2](#) were therefore selected because each addresses a possible threat to the warrant linking item performance to listening ability. Excessive passage-key or stem-key overlap, for example, may allow candidates to answer through lexical matching rather than comprehension. Implausible or ungrounded distractors may make the correct answer identifiable without adequate processing of the lecture. Similarly, longest-correct-option bias, uneven answer-key distribution, or option-position effects may introduce construct-irrelevant variance by encouraging strategic responding rather than listening-based

Table 2

Quality-control indices and their implications for validity in GPT-generated academic listening items.

Quality control index	Brief definition	Implications for validity
Answer key cue/clue detection	Passage-to-key and stem-to-key lexical and semantic overlap	Invalidates the item since the response process would not rely on listening comprehension but on a cue internal to the item.
Distractor grounding	Degree to which distractors are based on information in the passage rather than invented, unrelated, or unsupported content	Weakens validity if distractors can be rejected without reference to the listening text. Well-grounded distractors help ensure that incorrect responses reflect plausible miscomprehension rather than poor item construction.
Longest-correct-option bias	Tendency for the longest option to be the correct one	Introduces construct-irrelevant variance by allowing test-wise candidates to identify the key through option length rather than comprehension of the lecture.
Answer key distribution	Frequency of correct answers across options A-D	A skewed distribution may create response-pattern cues and increase guessing effects. Balanced key placement helps reduce strategic responding unrelated to listening ability.
Question type distribution	Proportion of items tagged under each functional category	Supports construct representation by ensuring that the item set samples the intended listening subskills, such as gist, detail, inference, speaker attitude, and discourse organization.
Sentence complexity	Average sentence length and average word length in stems and options	Excessive linguistic complexity may shift the construct from listening comprehension to reading load, working memory, or vocabulary knowledge. Appropriate complexity helps keep the item focused on the targeted listening process.
Distractor plausibility	Plausibility, logical coherence, and specificity of distractors	Implausible distractors reduce item discrimination and weaken the warrant that selecting the key reflects listening comprehension. Plausible distractors make incorrect options attractive to candidates with partial or inaccurate understanding.
World knowledge reliance	Whether items can be answered without reference to the passage	Threatens construct validity when candidates can answer from background knowledge alone. Items should require information made available in the lecture so that performance reflects listening-based evidence.
Necessary information	Spans of the lecture transcript required to answer each item	Supports passage dependence and evidentiary alignment by showing that each item is answerable from identifiable information in the listening input. Missing or diffuse evidence weakens the link between the task and the construct.
Option validation	Whether the designated answer key is accurate and no alternative option could also be considered correct	Directly affects score interpretation. Ambiguous keys, multiple defensible answers, or inaccurate options undermine the inference that item responses provide valid evidence of the targeted listening ability.

reasoning. These concerns are also consistent with item-writing principles that emphasize clear stems, defensible keys, plausible distractors, and the avoidance of irrelevant cues in selected-response formats (Haladyna, 2022). Table 2 summarizes how each quality-control index was interpreted as evidence bearing on construct validity.

4. Results

4.1. Tier 1 results: rule-based methods

As previously discussed, a total of 40 listening tests comprising 240 items were generated across the three prompting iterations and the fine-tuned model. Next, we developed a two-tiered evaluation framework, which was applied to assess the quality of these L2 listening tests generated. In this section, we present the evaluation of the tests generated based on tier-1 or rule-based methods, consisting of test-level analysis, risk-tiering, and corpus-level analysis.

4.1.1. Test-level analysis

The test-level analysis comprises three analytical components: key cue/clue detection, weak distractor grounding, and binomial testing for longest-correct-option bias. For clarity, the results for each analytical component are presented separately for the three prompting iterations (Iterations 1, 2, and 3) and for the fine-tuned model.

Key Cue/Clue Detection. A series of lexical and semantic similarity analysis revealed minimal evidence of cueing or clueing across all three iterations. For each condition, both passage-to-key and stem-to-key overlap were computed using n-gram and sentence-embedding similarity measures. In Iteration 1, only one item (1.7%) exceeded both the N-gram and semantic similarity thresholds at the stem-to-key level, while all other comparisons fell below the cutoff. No instances of excessive lexical or semantic overlap were observed in Iterations 2, 3, or the fine-tuned condition. These results suggest that the prompts and fine-tuning processes effectively minimised unintended lexical similarity between items and their source materials, thereby reducing the likelihood of cueing effects.

Weak Distractor Grounding. To evaluate distractor plausibility, each distractor was assessed for its lexical and semantic grounding in the passage. In Iterations 1 and 2, 11 of 60 items (18.3%) contained at least one weakly grounded distractor. This proportion decreased to 8.3% in Iteration 3 and further to 3.3% in the fine-tuned condition. The progressive decline indicates that iterative prompt refinement and fine-tuning substantially improved distractor plausibility, reducing the prevalence of implausible or poorly grounded options.

Binomial Test for Longest-Correct-Option Bias. A binomial test was used to examine whether correct answers were disproportionately the longest options compared to chance expectation. Across all iterations, the observed proportions of longest-correct items significantly exceeded this baseline. In Iteration 1, a total of 39 of 57 items yielded a correct option that was the longest ($p = 6.79 \times 10^{-12}$), while Iteration 2 showed an even stronger bias with 50 of 57 items ($p = 2.92 \times 10^{-23}$). Iteration 3 similarly demonstrated a significant overrepresentation of longest-correct options (45 of 56; $p = 5.52 \times 10^{-18}$). Although the fine-tuned model exhibited a noticeably reduced magnitude of bias (35 of 55; $p = 1.66 \times 10^{-9}$), the effect remained statistically significant. The total number of items per iteration was slightly below 60 because items without a unique longest option were excluded from analysis. These results indicate that while fine-tuning mitigated the tendency toward longest-correct-option bias, it did not eliminate it entirely.

4.1.2. Risk analysis

Iteration 2 generated the highest risk score of 35 (Table 3), indicating that the refinements introduced at this stage did not resolve the underlying issues and, in some cases, exacerbated them. Although key cueing/clue detection and distractor grounding continued to improve marginally, the test-level longest-correct-option bias worsened

Table 3

Results of item-level analyses for materials generated through prompting and fine-tuning (rule-based results or tier 1).

Frequency and severity of item flaws across iterations (These were used to derive cumulative risk scores.)				
	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration
Key Cueing/Clue Detection	3	2	1	1
Weak Distractor	18	15	12	6
Longest-Correct-Option Bias	10	18	18	10
Total	31	35	31	17
Test counts across risk levels ^a and iterations				
	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration
High	9	9	9	5
Medium	1	0	0	2
Low	0	1	1	3
Total	10	10	10	10
Answer (key) distribution across iterations				
	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration
A	6 (10.0)	9 (15.0)	46 (76.7)	11 (18.3)
B	33 (55.0)	34 (56.7)	10 (16.7)	30 (50.0)
C	20 (33.3)	17 (28.3)	4 (6.7)	18 (30.0)
D	1 (1.7)	0 (0.0)	0 (0.0)	1 (1.7)
Question tag (subskill) distribution across iterations				
	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration
Detail	15 (25.0)	12 (20.0)	12 (20.0)	15 (25.0)
Gist-Content	10 (16.7)	10 (16.7)	10 (16.7)	9 (15.0)
Inference	9 (15.0)	8 (13.3)	7 (11.7)	7 (11.7)
Attitude	8 (13.3)	9 (15.0)	7 (11.7)	6 (10.0)
Function	7 (11.7)	8 (13.3)	12 (20.0)	13 (21.7)
Organisation	6 (10.0)	9 (15.0)	7 (11.7)	8 (13.3)
Gist-Purpose	4 (6.7)	2 (3.3)	4 (6.7)	2 (3.3)
Connecting Content	1 (1.7)	2 (3.3)	1 (1.7)	0 (0.0)

^a Lower risk levels are desirable.

markedly (Table 3). As in Iteration 1, nine of ten tests were flagged as High risk. One test was of Low risk, hinting at some improvement.

Iteration 3 showed little improvement overall, with the risk score again totalling 31 (Table 3), as was in Iteration 1. However, key cueing/clue detection and distractor grounding had shown an overall downward trend across iterations (Table 3). Longest-correct-option bias showed no improvement. Importantly, although the cue/clue rate decreased in some tests, these reductions were insufficient to shift overall risk classifications, and the same nine tests remained categorised as High risk, while the remaining test was of Low risk.

By contrast, the fine-tuned model exhibited the clearest reduction in risk, with a total score of 17 (Table 3), representing nearly a 50% decrease relative to Iteration 3 with the base model. Notably, both the number of longest-correct-option bias flags as well as key cueing/clue detection and weak distractor grounding were almost halved (Table 3). Five tests were classified as Low or Medium risk, a result not observed in earlier prompt-only iterations. Nonetheless, none of the risk factors were fully eliminated, indicating that fine-tuning delivered improvements in consistency and alignment but did not fully resolve all structural flaws.

4.1.3. Corpus-level analysis

Answer Key Distribution The distribution of correct answer keys showed varying degrees of imbalance across iterations relative to the

ideal uniform distribution of 25% per option (Table 3). Iterations 1 and 2 exhibited skew, with key B selected more than twice as often as expected (55–57%), while keys A and C were underrepresented and key D was almost not represented. Iteration 3 displayed substantial imbalance, with key A accounting for 76.7% of all correct answers. Similar to the previous iterations, keys A and C were underrepresented and key D was not represented. Following fine-tuning, the distribution became relatively more even, though some deviation persisted, with key B still overrepresented (50%) and the keys A and C closer to target proportions, while key D was still not represented. Overall, fine-tuning reduced key frequency bias but did not completely eliminate key issues.

Question Tag Distribution. The distribution of question types eliciting different listening subskills remained broadly consistent across iterations, with Detail and Gist-Content questions accounting for the largest proportions (15–25% and 15–17%, respectively). The imbalanced mix of question tags is acceptable as certain question types are naturally more frequently occurring than others.

Sentence Complexity. Basic linguistic measures were calculated to

Table 4

Basic item sentence complexity across iterations, broken down by question type.

Component	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration
<i>Average Words per Sentence</i>				
Stems	10.45	11.05	10.97	11.02
Attitude	11.75	12.33	12	13
Connecting Content	13	11	12	NA ^a
Detail	11.67	11.67	12.67	12.53
Function	10.57	12	11.33	11.08
Gist – Content	8.3	8.5	8.7	9.11
Gist – Purpose	9	8	8.75	9
Inference	11.89	12.75	11.86	11.57
Organisation	7.5	10.11	9.86	8.75
Options	6.06	7.33	8.85	9.08
Attitude	4.78	6.78	9.04	9.29
Connecting Content	7	7.25	8.5	NA ^a
Detail	5.5	7	8.5	7.8
Function	7.25	7.5	8.75	9.4
Gist – Content	5.45	7.17	8.55	9.17
Gist – Purpose	6.81	7.75	8.19	10.88
Inference	6.64	7.34	8.32	8.75
Organisation	7.25	8.28	10.86	10.56
<i>Average Word Length (in Characters)</i>				
Stems	5.07	5.29	5.31	5.34
Attitude	5.3	5.61	5.49	5.58
Connecting Content	4.85	5.68	6	NA ^a
Detail	5.2	5.43	5.38	5.43
Function	5.36	5.3	5.63	5.65
Gist – Content	4.05	4.22	4.41	4.49
Gist – Purpose	4.72	4.69	4.6	5.44
Inference	5.2	5.29	5.43	5.06
Organisation	5.44	5.67	5.51	5.5
Options	5.59	5.71	5.66	5.63
Attitude	6.04	5.97	5.77	5.73
Connecting Content	5.64	5.91	5.62	NA ^a
Detail	5.28	5.42	5.6	5.66
Function	5.13	5.49	5.43	5.41
Gist – Content	5.77	5.93	5.73	5.6
Gist – Purpose	5.83	5.92	5.6	5.52
Inference	5.36	5.75	5.82	5.66
Organisation	6.25	5.65	5.85	5.88

^a No “Connecting Content”-type questions were generated.

assess variation and readability across items (Table 4). Average sentence length and word length remained relatively stable across iterations, with slight increases observed following fine-tuning. For stems, average sentence length rose marginally from 10.45 to 11.02 words, while mean word length increased from 5.07 to 5.34 characters. For options, average sentence length showed a more pronounced increase, from 6.06 to 9.08 words, with word length remaining around 5.6 characters. These results indicate a gradual trend toward greater syntactic variety and lexical richness in the generated options, particularly after fine-tuning.

When disaggregated by question type, sentence complexity was broadly consistent across item categories, though a few patterns are worth noting. Gist-Content stems were consistently shorter than other item types across all conditions (ranging from 8.3 to 9.11 words per sentence), suggesting the more concise framing typical of gist-level questions. Option length showed the most pronounced increases following fine-tuning for Gist-Purpose and Organisation items, suggesting that fine-tuning encouraged more elaborated response options for these higher-order item types. Overall, however, variation in linguistic complexity across question types was modest, indicating that the model generated items of broadly comparable syntactic density regardless of the cognitive sub-skill being targeted.

Metadata Requirements. Finally, all tests generated contained exactly six items with four options (A–D), as required by the user prompt.

4.2. Tier 2 results: human evaluation

This section presents the results of the human evaluation (Tier 2 results), which examined the quality of the listening tests produced across the four experimental conditions. Inter-rater reliability statistics are reported in Section 3.5.1. Discrepancies were resolved through discussion, and all results reported here are based on the agreed-upon evaluations.

Distractor Issues Although distractor groundedness and plausibility worsened over the first three prompt engineering iterations, the fine-tuned model produced the best results, with the fewest distractors judged as ungrounded or implausible (97 and 20, respectively, out of 720 total distractors), as shown in Table 5. Notably, the decline in distractor quality across the initial three iterations contrasts with the improving trend observed in the rule-based evaluation. Possible reasons for this discrepancy are examined in Section 5.

In all four iterations, all distractors were found to be logical and appropriately specific.

World Knowledge Reliance The proportion of items answerable without access to the passage fluctuated across iterations, indicating varying degrees of world knowledge reliance (Table 5). Iterations 1 and 3 showed higher reliance ratios (0.47 and 0.53, respectively), suggesting that a greater share of items could be solved through background knowledge alone. In contrast, Iterations 2 and the fine-tuned condition demonstrated lower ratios (0.40 and 0.42), reflecting improved integration of passage-dependent information. Overall, fine-tuning reduced but did not completely eliminate the model's tendency to generate items that could be answered without contextual grounding.

Necessary Information Span Mean span coverage ratios increased steadily across the first three iterations, from 0.45 in Iteration 1 to 0.51 in Iteration 3 (Table 5), before slightly declining to 0.49 in the fine-tuned condition. This upward trend indicates that successive prompt refinements improved the model's ability to anchor items more precisely to relevant portions of the passage. Although the slight decrease following fine-tuning suggests some variability in alignment, overall coverage remained higher than in the initial iterations, reflecting greater localisation of necessary information within the text.

Item Quality by Question Type To examine whether fine-tuning benefits were distributed equally across cognitive subskills, human evaluation results were disaggregated by question type (Table 5). The findings presented in Table 5 reveal meaningful variation across item

categories. Attitude items showed the clearest and most consistent gains following fine-tuning across multiple dimensions: ungrounded distractors fell from 19 in Iteration 3 to 10, implausible distractors dropped from 8 to 3, and world knowledge reliance decreased substantially from 85.7 to 100.0% across prompt-only iterations to 50.0% after fine-tuning. Function items similarly benefited, with implausible distractors falling to 0 following fine-tuning and world knowledge reliance declining to 23.1%, the lowest observed across all conditions for this item type. Organisation items also improved notably in distractor groundedness (from 16 to 22 ungrounded distractors across prompt iterations to 13 after fine-tuning), though world knowledge reliance remained somewhat variable. By contrast, Inference items demonstrated more limited and uneven improvement. While ungrounded distractors decreased from 20 in Iteration 3 to 8 following fine-tuning, world knowledge reliance remained persistently high at 57.1% — the highest among all item types in the fine-tuned condition — suggesting that fine-tuning did not resolve the tendency to generate inference items answerable without genuine engagement with the passage. Detail items showed comparatively modest gains overall, with ungrounded distractors and world knowledge reliance remaining relatively stable across conditions. These patterns indicate that fine-tuning benefits item quality unevenly across subskills, with higher-order item types involving discourse-level reasoning or genuine inferential processing remaining more resistant to improvement. This finding is consistent with the qualitative observations reported in Section 5.2 and underscores the need for more targeted model conditioning strategies for complex item types.

Human-Generated Baseline To contextualize the AI-generated results against professional assessment standards, Table 5 also presents Tier 2 evaluation results for the human-generated TOEFL-style items used as the fine-tuning dataset ($n = 245$ items; 735 distractors). Several patterns are worth noting. For Distractor Issues, the human-generated baseline showed notably higher proportions of ungrounded distractors (617 out of 735) and implausible distractors (161) compared to all AI conditions. These figures are counterintuitive at first glance and should be interpreted with caution, as they may reflect differences in how groundedness and plausibility manifest in professionally developed items relative to AI-generated ones. For instance, human-authored distractors may be deliberately constructed to be plausible from a test-taker's perspective without being closely anchored to specific passage wording, which could result in higher ungroundedness flags under the present rubric. These patterns warrant further investigation and are acknowledged as a limitation of the current benchmarking approach.

For Absolute Words, human-generated items outperformed all AI conditions, with only 8 items (3.3%) containing absolute terms compared to a range of 9–23 across prompt-only iterations and 13 following fine-tuning, suggesting that avoidance of absolute language is a dimension on which AI-generated items have not yet reached desirable standards. For Necessary Information Span, human-generated items showed a notably lower mean span coverage ratio (0.32) than all AI conditions (0.45–0.51), indicating that expert-generated items draw on more localised and precisely targeted passage content, a characteristic that may reflect greater intentionality in item construction.

For World Knowledge Reliance, the human-generated baseline ratio (0.49) fell within the range observed across AI conditions (0.40–0.53), suggesting that passage dependence is a dimension on which the fine-tuned model (0.42) performed comparably to or better than the baseline. These comparisons provide an indicative reference point for interpreting the AI-generated results and suggest that while fine-tuning considerably narrowed the gap with professional standards on several dimensions, meaningful differences remain, particularly in the avoidance of absolute language and the precision of passage targeting.

Option Validation Human raters verified that each item contained a single, unambiguous correct answer consistent with the lecture content and that all options were independent. In Iteration 2, one item was judged invalid because the distractors were underdeveloped, making one of the distractors a plausible alternative answer. In Iteration 3, two

items were identified where the designated key could not be located in the passage or did not fully address the question. All other items across iterations contained valid answer keys. In addition, Iteration 1 contained one item with partially overlapping options (A: Carbon dioxide and glucose; B: Water and oxygen; C: ATP and NADPH; D: Oxygen and glucose), whereas all other items exhibited full option independence.

General Quality For general item quality, an increasing number of absolute terms were detected within distractors across the prompt-only iterations 1 to 3 (Table 5). However, this number decreased following fine-tuning, although it remained slightly higher than in the first iteration. No unnecessary negations, window dressing, trick or misleading questions, or opinion-based items were observed in any iteration.

Corpus-Level Analyses At the corpus level, all generated items were categorised under the correct question type, and topic, lecture type, and overall output structure were consistently aligned with their respective source passages.

4.3. Summary of findings

Across the three prompt iterations, improvements were observed in some dimensions but not others. Cueing and clueing issues progressively decreased, and distractor plausibility

Improved modestly over successive iterations. However, the longest-correct-option bias persisted, and in some cases worsened, while answer-key distributions remained uneven. Notably, the feedback from each iteration, as seen in Table 6, directly informed the refinement of the subsequent prompt, allowing for targeted adjustments based on observed weaknesses. Overall, these results indicate that while iterative prompt refinement enhanced certain aspects of item quality, it did not yield uniform gains across all dimensions.

Table 5
Human evaluation summary (or tier 2) across iterations, broken down by question type.

Metric	Prompting iteration 1	Prompting iteration 2	Prompting iteration 3	Fine-tuned iteration	Human-generated baseline
<i>Distractor Issues (Option Count)</i>					
Total distractor count	180	180	180	180	735
Not Grounded	121	129	146	97	617
Gist-Content	19	22	19	15	68
Gist-Purpose	6	4	10	1	19
Detail	26	25	27	24	147
Function	15	18	30	26	121
Attitude	19	22	19	10	30
Inference	18	12	20	8	160
Organisation	16	22	18	13	3
Connecting Content	2	4	3	NA ^a	69
Implausible	23	34	37	20	161
Gist-Content	0	0	3	2	5
Gist-Purpose	1	1	0	0	2
Detail	1	5	7	7	32
Function	0	2	8	0	35
Attitude	8	12	8	3	15
Inference	9	5	7	4	51
Organisation	2	7	4	4	1
Connecting Content	2	2	0	NA ^a	20
<i>Absolute Words (Item Count)</i>					
Total question count	60	60	60	60	245
Absolute Words	9	15	23	13	8
Gist-Content	0	0	2	3	0
Gist-Purpose	0	0	1	0	0
Detail	2	2	4	3	0
Function	0	0	5	0	1
Attitude	1	2	3	2	3
Inference	4	6	5	4	1
Organisation	1	4	3	1	0
Connecting Content	1	1	0	NA ^a	3
<i>World Knowledge Reliance (Ratio)</i>					
Can Answer Without Passage	0.47	0.40	0.53	0.42	0.49
Gist-Content	0.20	0.30	0.60	0.44	0.53
Gist-Purpose	0.50	0	0.75	1.00	0.64
Detail	0.40	0.17	0.42	0.47	0.43
Function	0.29	0.38	0.50	0.23	0.44
Attitude	0.88	1.00	0.86	0.50	0.40
Inference	0.56	0.63	0.43	0.57	0.56
Organisation	0.50	0.22	0.29	0.25	0.67
Connecting Content	1.00	0	1.00	NA ^a	0.50
<i>Necessary Information Span (Ratio)</i>					
Mean Span Coverage	0.45	0.47	0.51	0.49	0.32
Gist-Content	0.45	0.47	0.51	0.49	0.33
Gist-Purpose	0.43	0.54	0.50	0.50	0.27
Detail	0.48	0.50	0.51	0.50	0.31
Function	0.45	0.45	0.51	0.48	0.32
Attitude	0.47	0.49	0.54	0.51	0.28
Inference	0.42	0.44	0.50	0.46	0.31
Organisation	0.43	0.46	0.53	0.50	0.25
Connecting Content	0.57	0.44	0.55	NA ^a	0.36

^a No “Connecting Content”-type questions were generated.

Table 6

Prompt iterations and observation summary.

Iteration	Observation (finding)
1	• High incidence of longest-correct-option bias. • Frequent cue/clue matches (keys resembled passage/stem). • Distractors often weak and ungrounded.
2	• Answer-key distribution heavily uneven (A–D). • Longest-correct-option bias worsened. • Cue/clue problems reduced; distractors somewhat more plausible. • Very slight improvement in answer-key distribution.
3	• Cue/clue rates improved further; distractors stronger. • Longest-correct-option bias unchanged. • Answer-key distribution worsened.

5. Discussion

5.1. RQ1: effect of iterative prompt engineering on item quality and performance plateau

5.1.1. Selective improvements and trade-offs in iterative prompt engineering

This study set out to investigate whether fine-tuning provides a meaningful performance advantage over prompting for automatic item generation in L2 listening assessment. A total of four model conditions were examined: three prompt-only iterations and one fine-tuned model. Across these conditions, 40 listening tests comprising 240 multiple-choice items were generated and evaluated. A holistic assessment across all evaluation dimensions revealed that iterative prompt engineering led to selective improvements rather than uniform gains. Certain aspects, such as key cueing and clue detection, improved over successive iterations, whereas others—such as the even distribution of key letters and the frequency of absolute terms in distractors—deteriorated. This pattern suggests that prompt refinement can enhance one subset of item-quality attributes only at the expense of others. Such trade-offs likely stem from the opaque and highly entangled nature of LLM behaviour, where small changes to prompt phrasing or sequencing can trigger unpredictable shifts in generation patterns due to the model's black-box architecture (Chang et al., 2024). In practice, “mega-prompts”, like those used in iterations 2 and 3, containing extensive instructions may overwhelm the model's instruction-following capacity, leading to partial or inconsistent adherence across dimensions, while more concise prompts that prioritise a limited set of clearly defined objectives may instead yield outputs that are both higher quality and more stable (Levy et al., 2024). These findings collectively suggest that, beyond a certain point, the effectiveness of prompt engineering plateaus: additional detail does not guarantee improvement and may instead diffuse the model's focus across competing objectives.

5.1.2. Unpredictability of LLM behaviour

Furthermore, fluctuating results across dimensions such as world knowledge reliance, necessary information span coverage, and longest-correct-option bias illustrate the opaque and often unpredictable nature of large language model behaviour. Despite systematic prompt refinements, several dimensions appeared to improve or worsen independently of prompt adjustments, suggesting that prompt modifications do not always guide model outputs in the intended direction. This unpredictability reflects a broader limitation of current LLMs: their black-box architecture prevents direct interpretability or control over internal decision pathways. As such, prompt engineering functions as both a science and an art, requiring iterative experimentation and intuition rather than purely deterministic procedures. Moreover, the non-deterministic nature of LLM generation means that identical prompts can yield divergent outputs across runs (Chen et al., 2025), further complicating reliable evaluation. These stochastic effects are difficult to disentangle from genuine prompt-induced improvements unless sufficiently large sample sizes are produced at each iteration. Consequently, without large-scale sampling (e.g., exceeding 30

generated tasks per iteration), it remains challenging to distinguish meaningful prompt effects from random variance inherent in the model's generation process.

5.1.3. Divergence between rule-based and human evaluation

A notable discrepancy emerged between the rule-based and human evaluations of distractor groundedness. The rule-based analysis, which measured lexical and semantic similarity between distractors and the passage, suggested that distractors became increasingly grounded across prompt iterations. In contrast, human raters judged the opposite trend, identifying a growing number of ungrounded and implausible distractors. One explanation for this divergence lies in the differing operational definitions of “groundedness.” Rule-based metrics, which compute global similarity across the entire passage, may register high overlap even when a distractor is semantically related to peripheral or contextually irrelevant sections of the text. Human raters, however, tend to assess grounding locally—focusing on the region of the passage most directly associated with the correct answer—and may disregard distant or paraphrased references as insufficiently justified. A second factor concerns the covariance between implausibility and absolute wording: distractors containing absolute terms (e.g., always, never) were systematically marked as implausible by human raters, inflating perceived ungroundedness. Finally, LLM-generated distractors may exhibit surface-level lexical similarity without true semantic alignment. This phenomenon is known as shallow coherence and refers to the production of text that is locally fluent but lacks deeper semantic or discourse-level integration (de Wynter & Yuan, 2025). Taken together, these findings suggest that rule-based measures, while efficient, can overestimate semantic grounding by capturing lexical proximity rather than contextual relevance, underscoring the importance of human validation in evaluating meaning-based constructs such as plausibility and content grounding.

5.1.4. Additional insight from human evaluation

In addition to the quantitative results, human evaluation revealed several item-quality issues not captured by existing evaluation dimensions. First, some items labelled as inference questions did not require actual inferential reasoning. In such cases, the supposedly implied information was already stated explicitly in the passage. For example, a question asking “What can be inferred about the importance of photosynthesis for life on Earth?” paired with the key “The process is vital for both plants and animals” actually paraphrased the sentence from the passage, “this process is not only essential for plants, but for all living organisms” thus requiring recognition rather than inference. It is also important to note that inference-making was treated as a single construct in this study, whereas prior research distinguishes multiple types of inferencing in listening comprehension (Aryadoust, 2017). Future work should therefore unpack inference-making into its constituent subtypes and consider fine-tuning models to more accurately represent these distinctions.

Similarly, attitude-type questions were often answerable through surface cues or general plausibility, rather than genuine interpretation of the speaker's stance, which makes the more suitable for low-ability test-takers. Finally, items targeting organisational understanding of the passage frequently exhibited relatively poor distractor design. For instance, in the question “How is the information about photosynthesis organised in the lecture?”, the key (“By moving from general concepts to specific processes”) could be easily guessed based on common sense, with all other options clearly implausible (“By discussing historical scientists first”, “By comparing it with cellular respiration throughout”, “By presenting only the end results before the processes”). Such issues highlight that certain higher-order item types—particularly those involving discourse-level reasoning—remain challenging for prompt-only LLMs to generate reliably, underscoring the need for more specialised evaluation dimensions sensitive to pragmatic and rhetorical structure.

Overall, the findings indicate that iterative prompt engineering can improve certain surface-level and structural dimensions of item quality, such as cue reduction and formatting consistency. However, its impact on deeper semantic and psychometric qualities remains limited. Performance gains plateau once prompts become overly complex, with subsequent refinements producing diminishing or even adverse effects. Thus, while prompt engineering can optimise item generation within a bounded range, it cannot reliably control or sustain improvement across all dimensions of assessment quality.

5.2. RQ2: benefit of fine-tuning over prompt-only generation

5.2.1. Fine-tuning as a more stable and systematic alternative

Building on the findings from RQ1, it is evident that while prompt engineering produced partial and inconsistent improvements, fine-tuning yielded more stable and systematic gains. This extends previous research on LLM for AIG (Moon et al., 2022; Moore et al., 2023; Raina et al., 2023). Specifically, dimensions that had deteriorated during the prompt-only iterations—such as the frequency of absolute terms and the balance of key letter distribution—showed marked recovery after fine-tuning. Other aspects, including longest-correct-option bias and distractor groundedness, achieved their best performance among all experimental conditions. Although fine-tuning remains an opaque process, its effects are more predictable and robust, guiding the model to generate items that are better aligned with the characteristics of the training data and less prone to erratic behavioural shifts. The greater consistency observed after fine-tuning can be attributed to its capacity to internalise task-specific regularities within the model's parameters, as previously outlined. This process reduces the model's dependence on surface-level prompt wording. As a result, fine-tuned models exhibit greater alignment with task requirements, producing outputs that are less susceptible to random variation and prompt phrasing effects. In the context of language assessment, this means that fine-tuning not only enhances output stability but also improves the model's sensitivity to domain-relevant constraints, such as item structure, difficulty balance, and content validity.

Qualitative analysis further supports the quantitative findings, showing notable improvements in the fine-tuned iteration across multiple dimensions of item quality. Inference questions, which previously restated information explicitly stated in the passage, increasingly required genuine inferential reasoning. Attitude questions, which were earlier prone to test-wise guessing due to obvious sentiment cues, featured more balanced and plausible distractors. Items targeting the organisation of the passage also became more coherent, often including at least one distractor of high quality. In addition, distractor plausibility improved markedly: for instance, gist-purpose questions became more specific in context—asking, for example, “What is the professor's primary purpose in discussing project-based learning?” rather than the more generic “What is the professor's main purpose in this discussion?”—resulting in options that were closer in meaning and therefore less transparent. Furthermore, several complete tests could no longer be answered through world knowledge alone, reflecting stronger contextual dependence. Finally, inter-item cueing was substantially reduced: unlike earlier iterations where subsequent stems occasionally revealed answers to preceding questions, the fine-tuned items demonstrated improved independence and cohesion.

Beyond empirical confirmation, the findings offer insight into when fine-tuning yields distinct advantages over prompt-only approaches in assessment contexts. While carefully designed prompts may potentially approximate professional item-writing conventions, language assessment tasks require the simultaneous satisfaction of multiple latent constraints, such as passage grounding, distractor plausibility, and construct alignment, that are difficult to specify exhaustively or enforce consistently through declarative instructions alone. Fine-tuning appears particularly advantageous under these conditions, as it allows such constraints to be internalised through exposure to representative

examples rather than repeatedly restated at inference time. In this sense, the benefits of fine-tuning emerge not merely from increased task familiarity, but from its capacity to stabilise model behaviour in domains where quality is defined by the coordinated interaction of multiple assessment-relevant criteria.

5.2.2. Persistent structural issues: longest-correct-option bias and key distribution

While fine-tuning substantially improved most dimensions of item quality, two persistent issues remained: longest-correct-option bias and uneven key letter distribution, which have not been reported in previous research (e.g., Moon et al., 2022; Moore et al., 2023; Raina et al., 2023). Longest-correct-option bias, in particular, is a critical concern because it directly affects response validity, yet it persisted—albeit at a reduced level—even after fine-tuning. These persistent issues can be traced to the fundamental way LLMs generate text. During generation, the model predicts one token at a time based on preceding context, without maintaining an explicit global representation of the entire output sequence. As a result, it lacks an internal mechanism for quantitative comparison—such as counting the number of words in each option or monitoring the relative lengths of distractors versus the key. Even when the prompt explicitly instructs the model to ensure that all options are of comparable length or to balance the distribution of answer keys, these constraints are treated as linguistic cues rather than verifiable conditions. The model may reproduce the phrasing of such instructions in its internal representation but cannot systematically check its compliance once generation has begun. This token-by-token sampling process therefore makes it inherently difficult for LLMs to identify or correct the longest-correct-option bias or to enforce even key distributions across multiple items. In essence, these dimensions require forms of self-monitoring and symbolic reasoning that extend beyond the model's probabilistic text prediction, explaining why such issues might persist even under fine-tuned conditions.

As such, addressing these challenges may require integration with external reasoning or verification mechanisms. For instance, tool calling could enable the model to execute discrete operations (e.g., counting token length or verifying option balance) before finalising outputs. Similarly, step-by-step or self-verification approaches, including recursive “LLM-as-a-judge” methods, could be used to evaluate and revise generated items iteratively. Such hybrid strategies represent a promising direction for mitigating systematic biases that remain resistant to improvement through fine-tuning alone.

5.2.3. Benchmarking against human-generated listening MCQs

The comparison with the human-generated TOEFL-style baseline provides a useful but methodologically cautious reference point for interpreting the quality of the GPT-generated items. From an ECD perspective, the central issue is whether they support the intended evidentiary link between task performance and academic listening subskills, rather than whether AI-generated items superficially resemble human-generated items (Mislevy et al., 2003). On this basis, the results suggest a mixed pattern. Fine-tuning narrowed the gap with human-authored items on some validity-relevant dimensions, particularly world-knowledge reliance, where the fine-tuned model performed within the expert range and slightly better than the human-generated baseline. This finding is important because passage dependence is a basic requirement for valid listening assessment: items that can be answered without the passage provide minimal or no evidence of listening ability (Field, 2013; Holzkecht et al., 2021; O'Grady, 2024).

At the same time, human-generated items showed clearer advantages in avoiding absolute language and in targeting more localized spans of necessary information, suggesting greater control over item-writing conventions and evidentiary precision. This aligns with the view that effective MCQ design requires careful management of cueing, option structure, and the relationship between the stem, options, and passage content (Haladyna, 2022). The unexpectedly high rates of ungrounded

and implausible distractors in the human-generated baseline should therefore not be interpreted straightforwardly as evidence of poorer quality. Rather, they may indicate a limitation of the present rubric: professionally written distractors may be plausible because they reflect likely test-taker misunderstandings or discourse-level inferences, even when they are not closely tied to explicit passage wording. This is especially relevant for inference, attitude, organization, and connecting-content items, which require more interpretive processing than detail items and are harder to evaluate through surface grounding alone (Aryadoust, 2017; Aryadoust & Luo, 2023; Buck, 2001; Field, 2013). Overall, the expert comparison indicates that fine-tuning can improve the validity profile of AIG outputs, but it also shows that human benchmarking in listening MCQ generation requires evaluation criteria sensitive to the different ways human-authored and AI-generated distractors operationalise passage dependence, plausibility, and construct representation.

5.2.4. Defining item quality in L2 listening assessment

A central consideration raised by the present findings concerns the definition of a “good” L2 listening item. In this study, item quality was not conceptualised solely as the absence of technical flaws, but as the extent to which generated items were in alignment with an evidence-centered design. Specifically, items were expected to (a) elicit targeted listening subskills (e.g., gist, inference, discourse organisation), (b) require passage-based comprehension rather than test-wise reasoning or world knowledge substitution, and (c) minimise potential construct-irrelevant content arising from structural biases or superficial cues.

While Tier 2 metrics functioned as safeguards against technical flaws, they also served a construct-protective role. For example, excessive passage-key overlap threatens evidential interpretability potentially by enabling lexical matching strategies, rather than by eliciting listening subskills, whereas weak implausible distractors reduce the cognitive demands necessary for inference. Thus, the evaluation framework was designed not merely as a quality assurance checklist, but as an operational instantiation of construct protection principles in language assessment.

In summary, fine-tuning produced measurable and consistent gains over the prompt-only approach, yielding items that were more contextually grounded, linguistically coherent, and psychometrically balanced. While certain systematic issues, such as longest-correct-option bias and uneven key distribution, persisted, fine-tuning markedly reduced erratic variation and aligned model behaviour more closely with task-specific expectations. Overall, these results indicate that fine-tuning enhances both the stability and validity of LLM-generated items, offering a more reliable foundation for automatic item generation in language assessment.

6. Limitations

Several limitations of this study should be acknowledged. First, despite efforts to control for prompt structure and model variability, the inherently non-deterministic nature of LLMs introduces a certain level of stochastic variation that may have influenced the results. Second, the fine-tuning dataset, while task-specific, may not have captured the full range of listening and content variation present in operational assessment contexts. We emphasize that prompting and fine-tuning must be grounded in a clear theoretical specification of the underlying listening constructs. Future research should move beyond specifying the intended subskills or cognitive processes, as in the present study (Aryadoust & Luo, 2023; Buck, 2001), and examine whether LLMs can be trained or systematically prompted to generate items targeting a wider range of subskills. Existing theoretical frameworks in L2 listening, such as those synthesised by Aryadoust and Luo (2023) and chapters in Wagner et al. (2024), provide construct taxonomies and definitions that could inform such targeted model conditioning. Leveraging these frameworks in the design of prompting and fine-tuning protocols is likely to yield more

construct-faithful and cognitively valid items.

Third, although the present study did not include a fully controlled comparison between AI-generated items and items authored by human test developers, an indicative human-generated baseline was incorporated by applying the Tier 2 evaluation rubric to the human-generated TOEFL-style items used as the fine-tuning dataset ($n = 245$ items). The results of this comparison are presented in Table 5 and discussed in Section 4.2. While this baseline provides a useful reference point for contextualising AI-generated item quality against professional assessment standards, its interpretation is constrained by the use of a different rater from those employed in the primary evaluation, limiting direct metric comparability across conditions. Furthermore, the findings demonstrate relative gains across AI-based conditions rather than establishing equivalence with professional human practice. Future research should extend the present design by incorporating a fully controlled benchmarking comparison — using the same raters and evaluation instrument across both AI-generated and human-authored items — to more rigorously assess the extent to which fine-tuned models approximate professional standards in operational L2 listening assessment contexts.

A further limitation concerns the use of lecture transcripts rather than fully realised audio input. Listening assessment theory emphasises that authentic listening involves processing time-bound acoustic signals, including prosodic, phonological, and pragmatic cues such as stress, rhythm, and intonation. By focusing on written transcripts, the present study excludes these auditory dimensions. As such, the generated tasks more closely approximate academic listening comprehension at the discourse level than full real-time auditory processing. However, it should also be noted that this operationalisation reflects a focus on higher-level comprehension subskills (e.g., gist, inference, discourse organisation) that are central to academic listening constructs and account for substantial variance in performance among adult L2 learners; prior research (e.g., Buck, 2001; Vandergrift & Baker, 2015) suggests that lower-level perceptual features such as stress and intonation, while important for processing, may contribute less to score discrimination at advanced proficiency levels. Nonetheless, the absence of acoustic cues limits the ecological validity of the tasks and constrains the generalisability of findings to audio-based listening assessment. Future work should integrate speech synthesis and prosodic modelling to align automatic item generation more fully with authentic listening constructs.

Finally, although the proposed evaluation framework provides a more structured basis for item quality assessment, further validation using a larger generated test pool and additional raters is necessary to confirm its reliability and scalability.

7. Conclusion

This study examined how LLMs can be leveraged for AIG in L2 listening assessment, particularly within academic lecture-based task formats, through iterative prompt engineering and supervised fine-tuning. The findings showed that while prompt engineering yielded selective improvements, its effects were inconsistent and plateaued beyond a certain level of complexity. Fine-tuning, by contrast, produced more stable and contextually grounded items, demonstrating clearer alignment with task-specific expectations. A hybrid evaluation framework integrating rule-based and human evaluation methods was also proposed, providing a more systematic approach to assessing item quality. Together, these contributions highlight both the promise and current limitations of LLM-based item generation, pointing toward the need for continued refinement through larger datasets, hybrid reasoning mechanisms, and expanded validation in operational assessment contexts.

Additionally, the present comparison between prompt-only and fine-tuned models should be interpreted within the broader landscape of AIG methodologies. Existing approaches include template-based generation,

retrieval-augmented generation (RAG), and multi-stage LLM pipelines that combine rule-based filtering with model inference. The present study focuses specifically on the comparison between model-level adaptation and inference-time prompting, while keeping other architectural components constant. In doing so, it does not position fine-tuning as a substitute for empirically supported AIG strategies, but rather as a complementary mechanism that can be layered onto template-based or hybrid pipelines. Future work may therefore explore how fine-tuning interacts with retrieval-based constraints or rule-based post-processing to further enhance construct alignment and quality control.

CRedit authorship contribution statement

Vahid Aryadoust: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Joann Wong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Open data

This study did not collect original data. All listening tests used in this study are existing publicly available resources; details of these listening tests are provided in the relevant sections of the paper.

Ethics statement

This study involves no human participants, no collection of personal data, and no interventions of any kind. Ethical approval was therefore not required.

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Declaration of competing interest

Authors have no conflict of interest to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.caeai.2026.100623>.

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